

Moose: Latent concept learning with reasoning-shortcut awareness in $\mathcal{EL}^{++}$

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Abstract. The OWL 2 EL profile is used in some of the largest production ontologies, including the Gene Ontology and SNOMED CT. Existing neuro-symbolic (NeSy) learning methods accept propositional theories or Datalog, and reasoning-shortcut (RS) awareness has not been investigated in ontology settings. We present MOOSE, a method that compiles an $\mathcal{EL}^{++}$ TBox and finite ABox to a Sentential Decision Diagram (SDD). The SDD acts as a differentiable weighted-model-counting layer, and we add closure clauses outside the $\mathcal{EL}^{++}$ profile on declared exhaustive families to overcome the limited expressivity of $\mathcal{EL}^{++}$ under partial supervision. We show termination, soundness, completeness, and polynomial intermediate sizes, and validate the proofs in Lean. We then define the first formal partial-supervision latent-concept-learning task over an OWL EL ontology, i.e., learning per-individual classifiers for latent concepts from observed ABox literals, and evaluate MOOSE on MNIST-with-ontology and Pizzaiolo. MOOSE improves over propositional-NeSy, fuzzy-logic, and ontology embedding baselines, and presents the first reasoning-shortcut analysis in an OWL EL setting.

Keywords: OWL 2 EL · neuro-symbolic learning · knowledge compilation · weighted model counting · reasoning shortcuts.

1 Introduction

The OWL 2 EL profile underpins some of the largest and most widely-used ontologies, such as the Gene Ontology with $\sim 50\text{K}$ classes and $\sim 8\text{M}$ annotations [3,49], SNOMED CT [22], and most of the OBO Foundry ontologies [45,25], because EL trades expressivity for polynomial-time reasoning on subsumption, instance checking, and consistency [5,26]. Neuro-symbolic (NeSy) learning combines neural perception with symbolic constraints; several families coexist. Fuzzy / t-norm

methods replace propositional truth with continuous operators on $[0, 1]$ [7]; embedding methods score entities and relations in a continuous geometry [36,15,16]; *knowledge-compilation* methods compile the constraint to an arithmetic circuit whose weighted model count (WMC) is differentiable in the inputs [39,51,43,1]. The knowledge-compilation family retains an exact WMC of the constraint as its training signal under partial supervision.

The task we address, *ABox-supervised latent concept learning*, takes the following form. Each training instance supplies one perceptual input per named individual (e.g. an image of a digit) together with a partial set of ABox literals over an *observable* signature; the truth values of atoms over a separate *latent* signature are never directly supervised. The objective is to learn per-individual classifiers for the truth values of latent concept atoms. A WMC layer over a circuit compiled from the ontology supplies the training signal: evaluated with the perception’s per-atom outputs and clamped on the observed evidence, the layer scores how compatible the perception’s predictions are with the ontology, and the gradient steers the perception toward configurations that the constraints admit. When several latent configurations satisfy the same evidence, no method can recover the true labels from supervision alone; per-atom-independent predictors fail to recognize this and collapse to high confidence on a single arbitrary configuration rather than spreading mass over the satisfying ones [39,41,29].

Existing knowledge-compilation NeSy frameworks accept propositional formulas, Datalog programs, or finite-domain rules [39,51,43,1], but not OWL ontologies directly. Two recent works target description logics: a domino-style reduction of $\mathcal{ALCI}$ to a probabilistic circuit [37] that treats the circuit as a regulariser under full label supervision, and a fuzzy approximation of $\mathcal{EL}^{++}$ via Goguen-implication semantics [53] that targets knowledge-base completion without exact WMC or compilation soundness; neither performs partial-supervision concept learning over OWL EL. Embedding methods such as ELEmbeddings and OWL2Vec* [36,15] map ontology structure to a continuous geometry (via geometric model-theoretic constraints or graph walks respectively), trading exact entailment for differentiability. A separate body of work on reasoning shortcuts (RS) [41,40,29,31,30,9] formally proves that conditional independence among predicted concepts is incompatible with RS-awareness, and develops mitigation strategies (BEARS ensembles, neuro-symbolic diffusion); these results are stated entirely in propositional or Datalog NeSy. OWL 2 EL underpins many production ontologies, so the RS phenomena documented in propositional settings translate directly into uncertainty-modeling failures in deployed ontology-driven systems. Translations of OWL EL into other formalisms are well established: it is Datalog-rewritable [33,11], and probabilistic [12,13] and fuzzy [53,8] variants have been studied at length. However, a rewriting yields a reasoning procedure rather than a learning signal, and the probabilistic and fuzzy variants relax or re-specify the model-theoretic semantics rather than compiling it.

Here we present MOOSE, a method that compiles an $\mathcal{EL}^{++}$ ontology and a finite ABox domain into a Sentential Decision Diagram [20,17] whose models coincide with ABox interpretations satisfying the entailments of the input ontology,

paired with a learning framework that uses the differentiable circuit as the sole supervision channel for latent concept atoms and provides the first RS analysis in an OWL EL setting. Our contributions are:

1. **End-to-end OWL 2 EL compilation with full proofs.** We compile the EL profile, including role chains $R_1 \circ R_2 \sqsubseteq S$ and role hierarchies (the constructs that distinguish SNOMED CT and the Gene Ontology from a propositional or $\mathcal{ALCI}$ setting), via ELK-style saturation [26] to an SDD supporting WMC. The four formal contributions are a verified SDD encoding (Theorem 1), the rational DISPONTE [44] distribution-semantics correspondence (Theorem 2), and SCC-compositional factorizations at the Sat and WMC levels (Theorems 3 and 4). All four are mechanized in Lean 4 (no MOOSE-specific axioms; Sections D and D.10), along with the supporting Lean libraries for ELK saturation, soundness/completeness on $\mathcal{EL}^{++}$ [26], and SDD knowledge compilation, which to our knowledge are the first such formalizations (Section D.1). The closure-augmented variant is correct (Theorem 7) and inference is linear in the SDD (Theorem 8).
2. **ABox-supervised latent concept learning under partial axiom observation.** We pose a learning task in which a subset of ABox literals over the observable signature is given as evidence (with truth values written `true` and `false`, so that they are not confused with the $\mathcal{EL}^{++}$ concepts $\top$ and $\perp$), and the goal is to learn per-individual classifiers for the truth values of latent concept atoms (role atoms are observed or marginalized, never classified), to our knowledge the first such formulation over an OWL EL ontology, generalizing DeepProbLog [39] and Semantic Loss [51] to description logic constraints with role chains and role hierarchies.
3. **Reasoning-shortcut analysis transposed to OWL EL.** We compute family-argmax accuracy, expected calibration error (ECE), and the RS-consistency rate under MOOSE and two plug-in mitigations (MOOSE+BEARS [40] and MOOSE+NeSyDM [30]), and isolate a calibration-vs-accuracy trade-off: BEARS leads on family-argmax accuracy via ensemble diversification on RS-suspect inputs, NeSyDM leads on calibration (ECE) under symbolic ambiguity.
4. **A benchmark suite.** A single MNIST-with-ontology benchmark covering three supervision regimes (atomic, relational, and role-chain), together with a fourth experiment on the Pizzaiolo synthetic-image dataset [10] that transfers the method to a real expert-authored OWL EL ontology.

2 Background and related work

2.1 Preliminaries: OWL EL, knowledge compilation, partial-supervision NeSy

The OWL 2 EL profile. $\mathcal{EL}^{++}$ [5] is the description logic basis of OWL 2 EL (syntax and semantics in Table 4). Concepts are built from atomic names, $\top$, $\perp$, conjunction $C \sqcap D$, and existential restriction $\exists R.C$; an ontology $\mathcal{O}$ contains GCIs

$C \sqsubseteq D$, role inclusions $R \sqsubseteq S$, and role chains $R_1 \circ R_2 \sqsubseteq S$, with subsumption, instance checking, and consistency decidable in time polynomial in $|\mathcal{O}|$. ELK [26] realizes this bound through a consequence-based saturation closure under ten completion rules [48]. $\mathcal{EL}^{++}$ is moreover Datalog-rewritable: consequence-based reasoning can be recast as the evaluation of a Datalog program [33,11], and Stage 2 of our pipeline (§3.2) uses precisely this view, treating the ELK saturation as a monotone Datalog program. Such a rewriting yields a reasoning procedure, not a differentiable one. Probabilistic $\mathcal{EL}^{++}$ [12,13,24] attaches weights via Bayesian-network encodings; fuzzy $\mathcal{EL}^{++}$ [53,8] replaces model-theoretic semantics with a t-norm-based approximation.

Knowledge compilation, WMC, and SDDs. For a propositional formula φ and literal weights $w(\ell) \geq 0$, the weighted model count is $\text{WMC}(\varphi; w) = \sum_{M \models \varphi} \prod_{\ell \in M} w(\ell)$ [14]. With arbitrary non-negative weights WMC is a generic algebraic quantity; it acquires a probabilistic reading precisely when the weights are *per-variable normalized*, i.e. $w(X) + w(\neg X) = 1$ for every variable X . Under that condition the weights specify a product distribution over truth assignments, and $\text{WMC}(\varphi; w)$ is exactly the probability that a sample from this distribution satisfies φ : it lies in $[0, 1]$ and $\text{WMC}(\varphi; w) + \text{WMC}(\neg\varphi; w) = 1$ [14]. MOOSE enforces this normalization by construction (§3.1, Table 1), and Theorem 2 establishes the resulting WMC/distribution-semantics identity for the compiled circuit. When the per-variable weights are produced by a neural network, WMC is differentiable in the network’s outputs. Knowledge compilation translates φ to a circuit whose structural properties (smoothness, decomposability, determinism) make WMC linear in circuit size and reduce probabilistic queries to circuit traversals [21]. A Sentential Decision Diagram (SDD) [20,17] is one such circuit and supports polynomial-time conjunction, disjunction, conditioning, and weighted model counting; exact definitions are in Section D, and Section C.3 shows a small example.

Knowledge-Compilation NeSy and partial supervision. A predictor composes a neural concept extractor $p_\theta(\mathbf{c} \mid \mathbf{x})$ with a symbolic constraint φ and trains by maximizing the marginal

$$p_\theta(\mathbf{y} \mid \mathbf{x}) = \sum_{\mathbf{c}} p_\theta(\mathbf{c} \mid \mathbf{x}) \mathbf{1}[\varphi(\mathbf{c}) = \mathbf{y}], \quad (1)$$

the standard objective across this family [39,51,1,43,30]. DeepProbLog [39] grounds φ as probabilistic Datalog with neural-network annotated facts; Semantic Loss [51] adds $-\log \text{WMC}(\varphi)$ on propositional constraints; Semantic Probabilistic Layers [1] combine exact probabilistic inference with logical constraints in a single tractable circuit; Scallop [38] compiles differentiable Datalog with provenance semirings; NeurASP [52] embeds neural perception inside answer-set programs; and A-NeSI [32] amortizes the intractable WMC marginal with a neural surrogate. A *partial-supervision* instance reveals labels for a subset of $\mathbf{y}$ -atoms only; the latent concept vector $\mathbf{c}$ is recovered through the WMC gradient (Definition 1).

None of the existing frameworks accepts an OWL ontology directly: the user must hand-translate existentials, role hierarchies, and role chains, losing the soundness, completeness, and polynomial-time saturation guarantees of consequence-based EL reasoning [26] in the process.

2.2 The independence assumption and reasoning shortcuts

Knowledge-compilation NeSy predictors almost universally factorize $p_\theta(\mathbf{c} \mid \mathbf{x}) = \prod_i p_\theta(c_i \mid \mathbf{x})$. Marconato et al. [41] show that this can attain optimal log-likelihood while learning unintended concept semantics, i.e., a *reasoning shortcut* (RS), and identify four root causes (knowledge structure, ground-truth concept distribution, objective, extractor architecture); RSBench [9] quantifies RS rates across NeSy architectures. [29,31] prove that the independence factorization is incompatible with *RS-awareness*: placing non-trivial probability mass on every constraint-consistent latent assignment. Two mitigations are available: BEARS [40] keeps the independent extractor and trains a K -encoder ensemble whose members commit to distinct shortcut explanations; NeSyDM [30] replaces independence with a masked discrete-diffusion concept distribution. Every published RS result is propositional, Datalog, or finite-domain relational; no OWL fragment has been studied through this lens. Related work on description logic compilation [37,53], refinement [19,4], and ontology embeddings [36,15,16] is reviewed in Section G.

3 Methods

3.1 Task and ground vocabulary

Given an $\mathcal{EL}^{++}$ ontology $\mathcal{O}$ and a finite ABox domain $\Delta = \{a_1, \dots, a_m\}$, the *ground vocabulary* is $\mathcal{V}_\mathcal{O}^\Delta := \{C(a) : C \in \text{Sig}_C, a \in \Delta\} \cup \{R(a, b) : R \in \text{Sig}_R, a, b \in \Delta\}$. The modeller partitions each signature into *observable* and *latent* names, $\text{Sig}_C = \text{Sig}_C^o \uplus \text{Sig}_C^\ell$ and $\text{Sig}_R = \text{Sig}_R^o \uplus \text{Sig}_R^\ell$. A training instance $(\Delta, \mathbf{x}, \mathbf{e})$ comprises a per-individual input tuple $\mathbf{x} = (x_a)_{a \in \Delta}$ with each $x_a \in \mathcal{X}$ and a partial set of ABox literals $\mathbf{e}$ over the observable ground vocabulary (Definition 1, Section B). We learn a weight-shared per-individual classifier $f_\theta : \mathcal{X} \rightarrow [0, 1]^{|\text{Sig}_C^\ell|}$ that approximates the marginal of each latent atom under $\mathcal{O} \cup \mathbf{e}$. The differentiable WMC layer evaluates $\text{WMC}(\alpha; w, \mathbf{e})$ under the literal-weight map w that assigns $f_\theta(x_a)$ to each latent concept atom $C(a)$, its observed truth value to each observed atom, $\frac{1}{2}$ to each unobserved observable and to each latent role atom, and $1 - w(X)$ to each negative literal $\neg X$, where X ranges over the ground atoms of $\mathcal{V}_\mathcal{O}^\Delta$ (Table 1). Because $w(X) + w(\neg X) = 1$ for every X by construction, the map is per-variable normalized: it defines a product distribution over the ground atoms whose per-atom marginal is w , and $\text{WMC}(\alpha; w, \mathbf{e})$ is the probability under this distribution that a sample satisfies α and the evidence $\mathbf{e}$ (§2.1; Theorem 2). This product form encodes the per-atom *independence* factorization $p_\theta(\mathbf{c} \mid \mathbf{x}) = \prod_i p_\theta(c_i \mid \mathbf{x})$ shared by knowledge-compilation NeSy predictors (§2.2). We make the assumption explicit here because it is the object of study rather than an

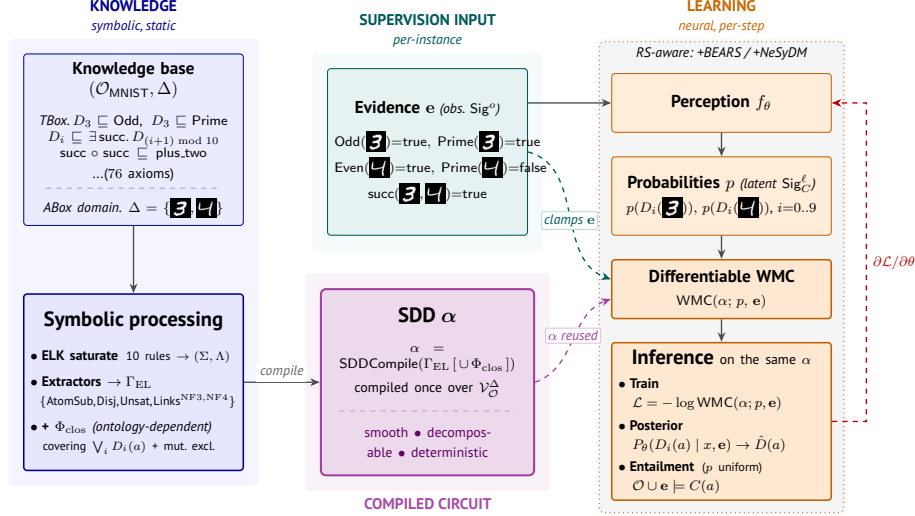


Fig. 1. End-to-end MOOSE workflow on $\mathcal{O}_{\text{MNIST}}$ at $\Delta=\{a, b\}$. The TBox is compiled once via ELK saturation $\rightarrow$ shape-aware extractors $\rightarrow \Gamma_{\text{EL}}$ (+ Φ_{clos} when supplied) $\rightarrow$ SDD α . The differentiable WMC layer evaluates $WMC(\alpha; p_\theta(\mathbf{x}), \mathbf{e})$ for the training loss, the conditional posterior, and the entailment query.

Table 1. Literal-weight map w ; the two weights of each variable sum to 1.

Literal	Weight	Literal	Weight
$C(a), C \in \text{Sig}_C^\ell$	$f_\theta(x_a)[C]$	$C(a), C \in \text{Sig}_C^o$ unobs.	$\frac{1}{2}$
$C(a) \in \mathbf{e}$	1	$R(a, b), R \in \text{Sig}_R^o$ obs.	1 or 0 from $\mathbf{e}$
$\neg C(a) \in \mathbf{e}$	0	$R(a, b)$ latent or unobs.	$\frac{1}{2}$
$\neg X$ for any atom X : $1 - \text{weight of } X$			

incidental choice: it is exactly the factorization that induces reasoning shortcuts, which contribution 3 quantifies via RS_{cons} and which the BEARS and NeSyDM variants (§3.4) are designed to mitigate.

3.2 Compilation algorithm

The algorithm (full pseudocode in Algorithm 1, Section C) compiles an $\mathcal{EL}^{++}$ ontology and a finite ABox domain to an SDD over $\mathcal{V}_O^\Delta$ in seven stages; the SDD is built once at startup and reused across every training step. Stage 1 runs the ELK saturation, which derives all entailed subsumptions and role links and records the grounded rule instances that justify each derivation. Stages 2–5 build a propositional encoding of the full saturation closure, including deeply nested existential derivations and cyclic dependency chains, via Clark completion, strongly-connected-component (SCC) detection, time-stamped unrolling, and conjunctive normal form (CNF) conversion. Stages 6–7 ground the saturation

onto the finite ABox domain and compile the resulting clauses into the SDD that the differentiable WMC layer evaluates.

The two groups of stages operate at different abstraction levels and serve complementary purposes. Stages 2–5 produce a TBox-level propositional theory whose variables are concept-level atoms such as $x_{C \sqsubseteq D}$ and $x_{E \xrightarrow{R} C}$, including derived complex concepts (e.g. $C \sqsubseteq \exists R.\exists S.D$); the input axiom variables α_i in this theory are free, so the theory supports *probabilistic ontology reasoning*: assigning a weight p_i to each α_i and computing the WMC gives $P(\text{query entailed})$. Stages 6–7 operate on the ground vocabulary $\mathcal{V}_{\mathcal{O}}^A$ whose variables are ABox atoms $C(a)$ and $R(a, b)$; for *NeSy learning* all ontology axioms are certain and the WMC literal weights come from a neural network’s per-atom outputs (Table 1), so the input-axiom tracking of Stages 2–5 is not consumed.

In the NeSy learning use case, Stages 6–7 return to the Stage 1 saturation output and extract the subset relevant for ABox grounding: subsumptions, disjointness, and role links between named concepts. Deeply nested existential derivations that Stages 2–5 process, including those that create cyclic SCCs requiring time-stamped unrolling, have no ABox-level counterpart: the complex concepts produced during saturation (e.g. $\exists R.C$ or $\exists R.\exists S.D$) do not lie in the named signature $\text{Sig}_C(\mathcal{O})$, and the shape-aware extractors of Stage 6 correctly filter them out. Stages 2–5 still serve a diagnostic and validation role: the Clark completion, SCC structure, and CNF size characterize the TBox complexity and verify that the Stage 6 filter captures every ABox-relevant consequence.

We summarize each stage below.

Stage 1: ELK saturation. $\text{Saturate}(\mathcal{O}, E)$ runs the ten ELK completion rules [26] and returns the saturation pair (Σ, A) , where Σ is the set of derived subsumptions $C \sqsubseteq D$ and A the set of derived role links $E \xrightarrow{R} C$, sound and complete with respect to $\mathcal{O}$ [26]. We write $\text{Sat}(\mathcal{O}) := \Sigma \cup A$ for the saturation closure. For each derived atom the saturation records all grounded rule instances that can derive it (there may be multiple justifications), and these form the input to Stage 2. When the ontology contains existential restrictions or role chains, the saturation may derive non-atomic concepts (e.g. $\exists R.C$, $\exists R.\exists S.D$); these are internal to the completion and are consumed by Stages 2–5 but filtered out by the shape-aware extractors of Stage 6. Algorithmic details are in Section C, and Section C.3 works one axiom end-to-end through the algorithm.

Stages 2–5: TBox-level encoding. The saturation of Stage 1 can be viewed as a monotone Datalog program [18]. Stages 2–5 turn it into a propositional TBox theory: Clark completion [18] replaces each derived atom’s justifications with a biconditional, Tarjan’s algorithm [47] isolates cyclic strongly-connected components, these are broken by SCC-local time-stamped unrolling, and the Tseitin transformation [50] yields an equisatisfiable CNF. With all input-axiom variables true this theory has $\text{Sat}(\mathcal{O})$ as its unique model; when the axioms carry weights it supports probabilistic ontology reasoning. For NeSy learning these stages are diagnostic: the deeply nested and cyclic derivations they handle

lie outside the named signature and are filtered by the Stage 6 extractors, so the ABox grounding of Stages 6–7 consumes only the Stage-1 saturation. The construction and its equations are given in Section C.

Stage 6: shape-aware extractors and ABox grounding. Stage 6 extractors return to the Stage 1 saturation output (Σ, A) and select the subset that maps to ground ABox atoms over $\mathcal{V}_{\mathcal{O}}^{\Delta}$. **AtomSub** emits $\neg A(a) \vee B(a)$ for every subsumption $A \sqsubseteq B$ in Σ where both A and B are named concepts in $\text{Sig}_C(\mathcal{O})$; **Disj** emits $\bigvee_i \neg A_i(a)$ for n -ary atomic disjointness $A_1 \sqcap \dots \sqcap A_n \sqsubseteq \perp$ (flattened from any parenthesization); **Unsat** emits $\neg A(a)$ for atomic $A \sqsubseteq \perp$; **Links** emits the NF3-forward clause $\neg E(a) \vee \neg R(a, b) \vee C(b)$ per saturation-derived atomic link $(E, R, C) \in A$ (i.e. $E \sqsubseteq \exists R.C$ with $E, C \in \text{Sig}_C(\mathcal{O})$), plus an NF4-reverse clause $\neg R(a, b) \vee \neg C(b) \vee E(a)$ when $L(E, R) := \{C : (E, R, C) \in A\}$ is a singleton (so the NF4 direction is deterministic). The union of the clauses produced by these extractors is the propositional Horn program Γ over $\mathcal{V}_{\mathcal{O}}^{\Delta}$; the saturation closure is polynomial by Theorem 6 and grounding multiplies it by at most $|\Delta|^2$ (Section D.4).

Stage 7: SDD compilation. Γ is compiled bottom-up into an SDD α that is smooth, decomposable, and deterministic [17,21]. The role-pinning caveat applies to the Links clause: ELK derives $E \sqsubseteq \exists R.C$ (existential, “some witness”); MOOSE encodes the propositional clause $E(a) \wedge R(a, b) \rightarrow C(b)$, which is universal on the named pair. Soundness depends on Δ being closed and on role atoms over $\text{Sig}_R^{\mathcal{O}}$ being pinned by evidence (a MOOSE-specific encoding choice for the finite-named-domain regime, not an ELK theorem). The differentiable WMC layer traverses α and evaluates $\text{WMC}(\alpha; w)$ by the standard smooth/decomposable/deterministic recursion (Equation (8) in Section C), with literal weights w read from Table 1. Decomposability factors the product at each decision node; determinism makes the prime/sub pairs mutually exclusive, so the sum is plain rather than an inclusion–exclusion. Conditioning on $\mathbf{e}$ clamps the literal weight of every observed atom to 0 or 1.

3.3 Closure axioms outside the EL profile

OWL EL is open-world and lacks both right-hand-side disjunction and cardinality constructors [5], so the covering axiom $\top \sqsubseteq D_0 \sqcup \dots \sqcup D_{K-1}$ on an exhaustive family is unstatable in EL. Under partial supervision the WMC gradient on unobserved members of such a family vanishes at every all-False extension (e.g. on MNIST, observing **Even**(a) without closure leaves the digit posterior uninformative). We therefore optionally append a finite set of *closure axioms* Φ_{clos} to Γ_{EL} on each modeller-declared exhaustive family $\mathcal{F}$: pairwise disjointness, the covering clause $\bigvee_i D_i(a)$, and (when each D_i carries a distinguishing property profile π_i) profile-keyed reverse implications. We write these three jointly manipulated components as $\Phi_{\text{clos}} = \Phi_{\text{mutex}} \cup \Phi_{\text{cover}} \cup \Phi_{\text{profile}}$. Two of the three are outside the EL profile but are standard OWL-DL machinery; MOOSE’s contribution is the ELK-via-SDD infrastructure that absorbs Φ_{clos} at the propositional level, so the same

compilation yields α_{EL} or α_{clos} with no method change. This absorption is free at the CNF level (Φ_{clos} adds at most $\binom{K}{2}m + (K+1)m$ clauses per declared family: $\binom{K}{2}m$ mutex, m covering, and up to Km profile-keyed reverse implications), but the polynomial saturation-closure bound of Theorem 6 applies to the EL Horn fragment only and does not transfer to $\Gamma_{\text{EL}} \cup \Phi_{\text{clos}}$: covering disjunctions can raise the primal-graph treewidth, and SDD compilation is worst-case exponential in $|I|$ with a $O(|I| \cdot 2^w)$ bound under treewidth w (Section D.5); inference remains $O(|\alpha|)$ in the SDD node count regardless (Theorem 8). Whether closure is needed is ontology- and regime-dependent: a forward-only EL hierarchy under factorized perception leaves the all-false latent assignment consistent with any property evidence, so the WMC gradient is uninformative and closure restores a non-trivial gradient (an architectural softmax head encoding mutex+covering, as DeepProbLog uses via $\text{nn}(\cdot)$, achieves the same effect by other means); when forward subsumption alone pins the latent vector, no closure is added. The choice is orthogonal to RS-awareness, which targets the cross-atom factorization at the perception output [41,29] and is implemented by the BEARS / NeSyDM wrappers of §3.4. Section B.1 summarizes the assumptions introduced by each pipeline component and what each concedes against open-world semantics.

3.4 Inference and RS-aware variants

The differentiable WMC layer supports three inference operations on the same SDD (training loss, conditional posterior, and entailment query), varying only literal weights and evidence. The training loss is $L(\theta; \mathbf{x}, \mathbf{e}) = -\log \text{WMC}(\alpha; p_\theta(\mathbf{x}), \mathbf{e})$ [51], which coincides with the DeepProbLog marginal loss on this Horn theory by Proposition 1. The conditional posterior $P_\theta(C(a) \mid \mathbf{x}, \mathbf{e})$ is the ratio of two WMCs with $C(a)$ clamped vs. left free; the perception-free entailment query $\mathcal{O} \cup \mathbf{e} \models C(a)$ holds iff $\text{WMC}(\alpha; w_{1/2}, \mathbf{e} \cup \{C(a) := \text{false}\}) = 0$ (the $\frac{1}{2}$ -weighted SDD acts as a satisfiability oracle). All three run in $O(|\alpha|)$ (Theorem 8, Equations (11) to (13) in Section E.1). The compilation algorithm is independent of the perception’s output distribution, so RS-aware methods (§2.2) plug in by replacing $p_\theta(\mathbf{x})$ in w without modifying α . MOOSE+BEARS [40] trains a K -encoder ensemble diversified by Kullback–Leibler (KL) divergence against the running ensemble average; MOOSE+NeSyDM [30] replaces $\prod_i p_\theta(c_i \mid \mathbf{x})$ with a masked discrete-diffusion q_θ , evaluated with both the REINFORCE leave-one-out (RLOO) estimator [28] and an exact-WMC gradient estimator. Wrapper losses with hyperparameters are in Section E.3.

4 Correctness and complexity

This section states the four theorems that constitute the formal contribution of MOOSE: a verified SDD encoding of the ELK-derived saturation, the unconditional rational DISPONTE correspondence, and the SCC-compositional factorizations at the Sat and WMC levels. Each theorem is mechanized in Lean 4; the infrastructure on which they rest (ELK soundness/completeness on $\mathcal{EL}^{++}$ and polynomial-time

Sat decidability [26]) is reused unchanged from the ELK literature and re-mechanized in our Lean library (Section D.1). Proofs, Lean theorem names, and audit-surface details are deferred to Sections D and D.10.

Notation. $\mathcal{O}$ ranges over $\mathcal{EL}^{++}$ ontologies (OWL 2 EL minus datatype properties and concrete domains). $\text{Sat}(\mathcal{O}, C, D)$ holds iff ELK derives $C \sqsubseteq D$ from $\mathcal{O}$ [26]. A world $M : \mathcal{O} \rightarrow \{0, 1\}$ selects the sub-ontology $\text{sel}(\mathcal{O}, M) := \{\alpha \in \mathcal{O} : M(\alpha) = 1\}$ (*sel* short for *selectedAxioms* in the Lean library). For a weight $w : \mathcal{O} \times \{0, 1\} \rightarrow \mathbb{Q}$, the rational DISPONTE marginal [44] is

$$\text{P}^{\mathbb{Q}}(\mathcal{O}, C, D, w) := \sum_{M : \mathcal{O} \rightarrow \{0, 1\}} \mathbf{1}[\text{Sat}(\text{sel}(\mathcal{O}, M), C, D)] \cdot \prod_{\alpha \in \mathcal{O}} w(\alpha, M(\alpha));$$

$\text{P}(\mathcal{O}, C, D, w)$ denotes the natural-valued analogue when w takes values in $\mathbb{N}$. $\text{cmp}(\mathcal{O}, C, D)$ (*cmp* short for *compileSat*) is the verified Shannon-tree SDD produced by the algorithm of §3.2, and $\text{WMC}^{\mathbb{Q}}(t, w)$ is its rational weighted model count.

Theorem 1 (Verified SDD encoding). *For every $\mathcal{EL}^{++}$ ontology $\mathcal{O}$ and concepts C, D , there exists an SDD tree t such that*

1. $\text{model}(t, M) \iff \text{Sat}(\text{sel}(\mathcal{O}, M), C, D)$ for every world $M : \mathcal{O} \rightarrow \{0, 1\}$;
2. for every weight w , $\text{WMC}(t, w) = \text{P}(\mathcal{O}, C, D, w)$;
3. $|t| = 2^{|\mathcal{O}|+1} - 1$.

Theorem 1 is the formal contract of MOOSE’s compilation step: the SDD’s models are exactly the worlds whose selected sub-ontology ELK-entails $C \sqsubseteq D$, the SDD’s WMC is the DISPONTE marginal under arbitrary weights, and the worst-case size is the explicit $2^{|\mathcal{O}|+1} - 1$ bound of the Shannon expansion (Section D.5). Lean proof: see Section D.

Theorem 2 (Rational DISPONTE correspondence). *For every $\mathcal{EL}^{++}$ ontology $\mathcal{O}$, every concepts C, D , and every rational weight w ,*

$$\text{WMC}^{\mathbb{Q}}(\text{cmp}(\mathcal{O}, C, D), w) = \text{P}^{\mathbb{Q}}(\mathcal{O}, C, D, w). \quad (2)$$

The identity holds unconditionally on rational weights: no distributional assumption is required. It is the formal warrant of the standard entailment-as-WMC-zero phrasing of probabilistic DLs [44] (at the uniform prior $w \equiv \frac{1}{2}$, $\text{WMC}^{\mathbb{Q}} = 0$ iff no world’s sub-ontology entails $C \sqsubseteq D$), and lifts that phrasing from the propositional encoding to the verified compiled circuit. Lean proof: see Section D.

Theorem 3 (SCC compositional theorem, Sat level). *Let $\mathcal{O} = \mathcal{O}_1 \uplus \mathcal{O}_2$ with disjoint signatures, both components nominal-free and range-chain-safe, $\mathcal{O}_2$ consistent ($\neg \text{Sat}(\mathcal{O}_2, \top, \perp)$), and C, D nominal-free with C, D in the signature of $\mathcal{O}_1$. Then*

$$\text{Sat}(\mathcal{O}_1 \uplus \mathcal{O}_2, C, D) \iff \text{Sat}(\mathcal{O}_1, C, D).$$

The result formalizes the intuition that an inferentially irrelevant SCC cannot affect Sat-derivability inside the relevant component, provided the irrelevant component is itself consistent. SCC-wise compilation follows: each SCC can be saturated and compiled in isolation, and the joint posterior recovered by combination. Range-chain safety rules out interactions between role chains and range axioms that would block the SCC factorization. It holds vacuously when neither component has range axioms (the case for both experimental ontologies); for chain-and-range ontologies, mechanized syntactic range-elimination [6] reduces to this case under a side condition that is automatically satisfied by reflexive role-inclusion propagation. Lean proofs, theorem names, and side-condition details: Sections D and D.10.

Theorem 4 (Per-SCC posterior equivalence). *Under the hypotheses of Theorem 3, for any uniform per-axiom prior $w \equiv c$ with $c > 0$, the per-SCC and joint DISPONTE posteriors coincide. In the unnormalized counting case $w \equiv 1$,*

$$P^Q(\mathcal{O}_1 \uplus \mathcal{O}_2, C, D, w) = P^Q(\mathcal{O}_1, C, D, w) \cdot 2^{|\mathcal{O}_2|}. \quad (3)$$

The multiplicative $2^{|\mathcal{O}_2|}$ from the irrelevant SCC cancels under posterior normalization, and the same cancellation extends the identity to any $c > 0$ after factoring out $c^{|\mathcal{O}_1|+|\mathcal{O}_2|}$. This is the WMC-level analogue of Theorem 3: per-SCC compilation is posterior-faithful. Lean proof: Sections D and D.10.

The closure-augmented variant of the SDD (which materializes the per-individual partition constraints on declared exhaustive families) satisfies an analogous correctness theorem (Theorem 7, appendix). Inference is linear in $|\text{cmp}(\mathcal{O}, C, D)|$ [17,21] (Theorem 8, appendix). All proofs and the DeepProbLog equivalence Proposition 1 are in Section D; the implementation index in Section D.10 cross-references every paper claim with its formalized counterpart.

5 Research questions and experiments

We evaluate MOOSE on two benchmarks: the MNIST-with-ontology benchmark we developed for this work (Example 1), instantiated under three supervision regimes (atomic property literals, relational $\exists R.C$ evidence, and role-chain evidence); and the Pizzaiolo dataset [10] of 4,800 synthetic pizza images generated to conform to the published OWL pizza ontology [46], which lets us test transfer to a third-party ontology and a different image distribution. Within each benchmark, the experiments share the ontology and differ only in the observable signature, the domain size $|\Delta|$, and the supplied evidence.

We address five research questions. **RQ1–RQ3** test latent digit recovery on $\mathcal{O}_{\text{MNIST}}$ as the supervision signal grows from a single individual with unary property literals (RQ1), to a pair of individuals connected by an observed $r(a, b)$ role literal under an $\exists R.C$ axiom (RQ2), to a pair connected by observed role literals only through an ELK-derived role-chain consequence (RQ3). **RQ4** tests transfer to the pre-existing third-party OWL pizza ontology, restricted

to its EL fragment (Pizzaiolo, §5.2). **RQ5** asks which RS-mitigation regime works under which conditions: with encoder, ontology, and compiled SDDs fixed, we vary only the output-distribution treatment: an independent per-atom baseline (no logical structure), MOOSE on the EL theory alone (the DPL surface, Proposition 1), MOOSE with closure Φ_{clos} , and the BEARS / NeSyDM wrappers, across factorized relational, tied symmetric, and high-arity disjunctive ambiguity. The independent baseline (binary cross-entropy, BCE, on observed atoms) is a lower-bound reference that cannot propagate evidence through subsumption, disjointness, or role axioms and so cannot beat chance on latent atoms [41,29].

5.1 The MNIST ontology and supervision regimes

A single OWL EL ontology $\mathcal{O}_{\text{MNIST}}$ (14 concepts, 2 roles, 76 axioms; full listing in Section F.2) is used in all three MNIST experiments; the experiments differ only in $|\Delta|$ and the evidence regime, never in the TBox. ELK saturation derives the chain consequence $D_i \sqsubseteq \exists \text{plus_two}. D_{(i+2) \bmod 10}$ from the axiom $\text{succ} \circ \text{succ} \sqsubseteq \text{plus_two}$ at compile time. The compiled SDD has $14m + 2m^2$ ground atoms. A small convolutional neural network (CNN) f_θ (2 conv blocks + 2 fully-connected layers, weight-shared across individuals; Section F.3) maps each image to per-atom probabilities; training minimizes $\mathcal{L}(\theta) = -\log \text{WMC}(\alpha_{\text{clos}}; p_\theta(\mathbf{x}), \mathbf{e})$ with Adam, batch 32, 5 seeds; per-regime learning rates and epoch counts are in Section F.4. EL-only ablations omit Φ_{clos} and train against α_{EL} , recovering the DPL loss (Proposition 1). **Exp. 1 (atomic, RQ1)** uses $\Delta = \{a\}$ with $n_{\text{obs}} \in \{1, 2, 3\}$ unary literals over $\text{Sig}_C^o = \{\text{Even}, \text{Odd}, \text{Prime}, \text{Composite}\}$. **Exp. 2 (relational, RQ2)** uses $\Delta = \{a, b\}$ with the role atom $\text{succ}(a, b)$ asserted (digits satisfy $\text{digit}(b) = (\text{digit}(a)+1) \bmod 10$) plus $n_{\text{obs}} \in \{1, 2, 3\}$ primality literals. **Exp. 3 (role chain, RQ3)** flips the asserted role to $\text{plus_two}(a, c)$ and widens the unary pool to parity and primality; the only chain from a to c is the NF7-derived consequence above. The digit family is declared exhaustive at every individual; Φ_{clos} supplies pairwise disjointness, covering, and reverse-implication clauses. Some property profiles uniquely identify the latent digit (e.g. $\text{Even}(a) \wedge \text{Prime}(a)$ pins D_2), while others leave it ambiguous (e.g. $\text{Even}(a)$ alone admits $\{D_0, D_2, D_4, D_6, D_8\}$); RS_{cons} on the ambiguous cases measures confident commitment to a wrong digit.

5.2 Experiment 4: Pizzaiolo (RQ4)

Pizzaiolo [10] provides 4,800 synthetic pizza images. The canonical pizza ontology contains non-EL constructs (universal restrictions on hasTopping , complement-based definitions such as $\text{VegetarianPizza} \equiv \text{Pizza} \sqcap \neg \exists \text{hasTopping.MeatTopping} \sqcap \neg \exists \text{hasTopping.FishTopping}$, and cardinality on NumberedPizza); we use only its EL-expressible fragment, restating the four-pizza recipes and property classes as conjunctive subsumption and disjointness axioms, and treating classes whose canonical definitions fall outside EL (e.g. VegetarianPizza) as atomic concepts whose truth values come from the dataset labels rather than from non-EL closure. The method targets the four-pizza subset

$\{\text{Mushroom}, \text{Cajun}, \text{Capricciosa}, \text{FourSeasons}\}$ over 16 topping concepts, with conjunctive per-pizza axioms (e.g. $\text{Capricciosa} \sqsubseteq \text{Ham} \sqcap \text{Anchovy} \sqcap \text{Olive} \sqcap \text{Peperonata}$), pairwise pizza disjointness, and disjointness axioms $P \sqcap T \sqsubseteq \perp$ for every topping T outside P 's recipe. EL forward subsumption plus these disjointness axioms pin the topping vector once pizza identity is observed; no out-of-profile closure is added. **Track A** reveals pizza identity; toppings are latent. We test distribution shift via an OOD split: the four training pizzas all carry **Anchovy** and **Olive** together, so a network can score well on the in-distribution split by coupling the two atoms; we hold out five tie-breaker pizzas (**Fiorentina**, **Giardiniera**, **LaReine**, **Soho**, **Veneziana**) whose recipes break this co-occurrence. **Track B** reveals one of four property-class atoms (**NonVegetarianPizza**, **SpicyPizza**, **RealFrenchPizza**, **VegetarianPizza**); revealing **NonVegetarianPizza** leaves $\{\text{Cajun}, \text{Capricciosa}, \text{FourSeasons}\}$ indistinguishable (a 3-way RS over pizza identity). **Track C** (the `is_spicy` task in the codebase) adds four axioms $T_i \sqsubseteq \text{SpicyTopping}$ for $T_i \in \{\text{Jalapeno}, \text{Peperonata}, \text{PeperoniSausage}, \text{Prawn}\}$ together with $\text{SpicyTopping} \equiv \text{SpicyPizza}$ (two GCIs; here **SpicyTopping** acts as a label-class denoting ‘‘pizza with a spicy topping’’). Symbolic supervision $\text{SpicyPizza}(a)$ forces $\text{SpicyTopping}(a)$ but the EL theory leaves the witness ambiguous over $2^4=16$ topping subsets (the multi-witness regime BEARS/NeSyDM target). We train the perception CNN from scratch with batch size 32, 5 seeds, and 60 epochs on Tracks A and B (Track C uses 15 epochs); per-method hyperparameters and the NeSyDM (γ_c, γ_h) sweep results are in Section F.4.

5.3 Baselines, metrics, and results

We compare MOOSE against seven baselines. **Independent** (BCE on observed atoms) is the NeSy-free lower bound. **Semantic Loss** [51] shares the $-\log \text{WMC}$ objective but compiles only the directly-stated NF1/NF2 atomic axioms, dropping NF3/NF4/NF7 and ELK saturation, isolating the EL-aware method of §3.2. **DeepProbLog** [39] hand-codes each MNIST regime as a ProbLog program with one `nn(.)` digit directive; by Proposition 1 DPL and MOOSE compute the same loss on the same Horn theory, though that directive is an annotated disjunction, so DPL is not closure-free. **Moose+BEARS** [40] keeps the factorized perception and α_{clos} but replaces the single encoder with $K=5$ diversified encoders. **Moose+NeSyDM** [30] replaces the factorized extractor with a masked-diffusion concept distribution; we evaluate both RLOO and exact-WMC gradient estimators. **LTN** [7] substitutes a fuzzy product T-norm; **ELEmbeddings** [36,54] embeds the EL ontology into ball geometries with a classifier head; both are detailed in Section E.4.

Metrics. The headline tables Tables 2 and 3 report the regime’s principal accuracy and ECE, the expected calibration error binned by predicted confidence [23]. The two ontologies do not admit the same accuracy metric. On MNIST the digit family is declared exhaustive and we report Acc_F , *family-argmax* accuracy: per individual, the argmax of the WMC posterior over that family. On Pizzaiolo we report Acc_C , per-atom accuracy on the latent slice, since Track C declares no exhaustive family and the decode degenerates on the other two (Section F.7). The

Table 2. Principal accuracy (% , mean $\pm$ s.d. over 5 seeds. MNIST Experiments 1–3 (Section 5.1), Pizzaiolo Tracks A–C (Section 5.2); Track A is the out-of-distribution split. Metric: Acc_F (family-argmax) on MNIST, Acc_C (per-atom, latent slice) on Pizzaiolo; compare only within a column (Section F.7). Operative theory: $\Gamma_{\text{EL}} + \Phi_{\text{clos}}$ for the MOOSE rows on MNIST, Γ_{EL} elsewhere, none for Independent (§3.3). Bold = column-best; † = highest but not significant (Section F.8); DeepProbLog on Pizzaiolo is not a separate run (Section F.6). Full Acc_{atom} , NLL, RS_{cons} , and F1_{macro} in Table 9.

Method	MNIST			Pizzaiolo		
	Exp 1	Exp 2	Exp 3	Pz A	Pz B	Pz C
Independent	13.2 $\pm$ 7.9	8.8 $\pm$ 4.5	12.2 $\pm$ 7.3	21.7 $\pm$ 9.7	25.0 $\pm$ 0.0	48.9 $\pm$ 5.6
DeepProbLog	42.1 $\pm$ 1.3	38.9 $\pm$ 3.2	59.6 $\pm$ 13.6	68.1 $\pm$ 0.3	84.5 $\pm$ 2.1	81.0 $\pm$ 3.0
LTN	25.9 $\pm$ 6.5	14.5 $\pm$ 4.9	11.0 $\pm$ 7.9	22.1 $\pm$ 5.2	32.5 $\pm$ 20.9	84.8 $\pm$ 0.4 †
ELEm/mOWL	34.9 $\pm$ 4.2	34.8 $\pm$ 3.6	27.4 $\pm$ 7.1	25.7 $\pm$ 7.3	25.0 $\pm$ 0.0	75.4 $\pm$ 6.6
MOOSE	48.1 $\pm$ 10.1	74.6 $\pm$ 0.3	96.1 $\pm$ 7.2	68.1 $\pm$ 0.3	84.5 $\pm$ 2.1	81.0 $\pm$ 3.0
+BEARS	50.1 $\pm$ 5.2 †	76.5 $\pm$ 5.9 †	89.2 $\pm$ 7.2	68.6 $\pm$ 0.3	87.9 $\pm$ 0.9	84.2 $\pm$ 3.6
+NeSyDM (RLOO)	46.6 $\pm$ 2.1	58.2 $\pm$ 1.4	62.7 $\pm$ 2.9	72.2 $\pm$ 1.3	78.5 $\pm$ 0.8	63.9 $\pm$ 3.7
+NeSyDM (exact)	49.4 $\pm$ 9.7	62.6 $\pm$ 6.1	60.1 $\pm$ 5.7	67.4 $\pm$ 1.4	84.7 $\pm$ 2.2	80.1 $\pm$ 1.5

Table 3. Expected calibration error ECE (% , mean $\pm$ s.d.). Lower is better. Same baselines, seed counts, operative theories, and metric conventions as Table 2; bold = column-best per regime, not significance-tested. Per-regime Acc_{atom} , NLL, RS_{cons} , and F1_{macro} alongside ECE are in Table 9 (Section F.6).

Method	MNIST			Pizzaiolo		
	Exp 1	Exp 2	Exp 3	Pz A	Pz B	Pz C
Independent	9.3 $\pm$ 4.5	13.4 $\pm$ 3.6	9.0 $\pm$ 8.2	58.3 $\pm$ 15.4	26.4 $\pm$ 17.1	22.1 $\pm$ 9.2
DeepProbLog	10.3 $\pm$ 0.9	12.3 $\pm$ 0.9	8.0 $\pm$ 2.9	30.8 $\pm$ 0.7	15.5 $\pm$ 2.1	15.3 $\pm$ 1.4
LTN	16.1 $\pm$ 6.7	10.0 $\pm$ 0.0	10.0 $\pm$ 0.0	52.7 $\pm$ 21.8	39.3 $\pm$ 17.4	14.1 $\pm$ 0.5
ELEm/mOWL	3.9 $\pm$ 0.4	7.2 $\pm$ 0.1	6.3 $\pm$ 0.1	49.7 $\pm$ 10.2	16.1 $\pm$ 0.2	11.8 $\pm$ 3.8
MOOSE	10.6 $\pm$ 1.8	4.9 $\pm$ 0.3	1.1 $\pm$ 1.5	30.8 $\pm$ 0.7	15.5 $\pm$ 2.1	15.3 $\pm$ 1.4
+BEARS	3.9 $\pm$ 1.0	4.0 $\pm$ 1.2	5.6 $\pm$ 1.1	28.2 $\pm$ 1.5	12.3 $\pm$ 1.0	11.3 $\pm$ 3.4
+NeSyDM (RLOO)	6.0 $\pm$ 0.6	7.3 $\pm$ 0.7	4.6 $\pm$ 0.8	7.5 $\pm$ 1.2	2.6 $\pm$ 0.9	8.9 $\pm$ 1.5
+NeSyDM (exact)	5.1 $\pm$ 1.5	7.4 $\pm$ 1.2	7.9 $\pm$ 1.1	30.8 $\pm$ 3.2	15.3 $\pm$ 2.2	6.9 $\pm$ 2.2

full per-regime breakdown (Table 9, Section F.6) additionally reports the negative log-likelihood (NLL), the mean per-atom Bernoulli loss on the latent signature; the RS-consistency rate $\text{RS}_{\text{cons}} = \text{Conf} \cdot (1 - \text{Acc}_F)$ on the latent family-argmax (higher = more confident commitment to a wrong shortcut; on Pizzaiolo the complement uses Acc_C , and on Track C it is not applicable); per-atom accuracy Acc_{atom} on the latent signature; and the macro-averaged per-concept F1, F1_{macro} . A paired significance analysis of the headline comparisons over 20 seeds is given in Section F.8, and a learning-rate sensitivity analysis in Section F.5.

Findings on Experiments 1–3 (RQ1–3, RQ5). At $|\Delta|=1$ (Exp. 1) MOOSE leads DeepProbLog on Acc_F (48.1 vs. 42.1). Independent stays near random ($\text{Acc}_F = 13.2$). BEARS cuts ECE from 10.6 to 3.9 at no cost to accuracy; at $|\Delta|=1$ there is little diversifiable structure left for the ensemble to exploit. NeSyDM matches MOOSE on Acc_F (46.6 and 49.4 for the RLOO and exact estimators vs. 48.1). On the relational and role-chain regimes (Exps. 2–3) MOOSE dominates DeepProbLog by tens of points on Acc_F (74.6 vs. 38.9 on Exp. 2; 96.1 vs. 59.6 on Exp. 3): the EL-aware Links extractor propagates evidence across the $|\Delta|=2$ ground individuals, but the closure clauses Φ_{clos} are what make this signal learnable.

The ablation in Section F.10 shows the base WMC objective falling to 17.4 and 9.4 (near chance) once Φ_{clos} is removed, so the EL compilation supplies the structure and Φ_{clos} the identifying constraint, with reasoning-shortcut mitigations partially substituting for the latter (Table 13). BEARS reduces RS_{cons} (Table 9, Section F.6) but gives up a few points of accuracy on Exp. 3 (89.2 vs. 96.1); NeSyDM lags even at tuned (γ_c, γ_h) . LTN and ELEmbeddings under-perform on Acc_F (14.5 and 34.8 on Exp. 2): LTN’s ECE saturates near the uniform prior (10.0, information-vacuous), while ELEmbeddings’ lower ECE on Exps. 2–3 (7.2, 6.3) does not translate into accuracy. An inductive held-out-edge split, in which query individuals appear only in relational configurations never supervised, leaves MOOSE’s accuracy essentially unchanged while NeSyDM collapses to near-chance (Section F.9).

Findings on Experiment 4 (RQ4–5). Track A is an OOD generalization test on five held-out tie-breaker pizzas whose recipes break the {Anchovy, Olive} co-occurrence admissible on the four training pizzas; on the in-distribution split most methods reach near-perfect topping accuracy, but on the OOD split the MOOSE family clusters around 67–72% on Acc_C (Table 2). Track B is the canonical RS test (3-way symbolic ambiguity: a property-class observation leaves three pizzas indistinguishable). Plain MOOSE-WMC commits to a single constraint-consistent latent (“Mechanism A” of [41]); BEARS lifts Acc_C from 84.5 to 87.9 via ensemble diversification on RS-suspect pizzas, and NeSyDM (RLOO) trades a few points of accuracy (78.5) for the lowest ECE of any row (2.6). Track C (is.spicy disjunction) is the multi-witness regime NeSyDM was designed for: among the MOOSE variants BEARS leads on Acc_C (84.2 vs. 81.0 for plain MOOSE-WMC) and NeSyDM (exact) leads on ECE (6.9, the column best); LTN matches BEARS on accuracy (84.8, the highest value in the column but not significantly so, Section F.8) at higher ECE (14.1). On both Track B and Track C the MOOSE-family results separate along an argmax-vs-calibration axis (BEARS leads accuracy, NeSyDM leads ECE). These mitigation effects are sizeable rather than marginal: paired over 20 seeds, BEARS significantly exceeds plain MOOSE-WMC on both Track B and Track C, and NeSyDM significantly exceeds it on the Track A OOD split (Section F.8). In the symbolically ambiguous regimes it is therefore the RS-mitigation wrapper, not the exact base layer, that is the stronger configuration; the mitigation is doing substantive work, not fine-tuning. This reverses the relational regime (Exps. 2–3), where forward subsumption already pins the latent vector, no residual reasoning shortcut remains to remove, and plain MOOSE is best; mitigation helps precisely where symbolic ambiguity leaves an RS to exploit. Whether this generalizes beyond the regimes tested here is left to future work.

6 Conclusion and outlook

MOOSE compiles OWL 2 EL into a Lean 4-verified weighted-model-counting layer with native role chains and role hierarchies, and supplies the first end-to-end formally verified circuit for an OWL profile together with the first reasoning-shortcut analysis in OWL EL. Empirically, the closure-augmented variant dominates propositional NeSy baselines by tens of points on relational and role-chain regimes, where the EL-aware grounding propagates evidence that propositional

encodings cannot and the closure clauses render the resulting signal learnable (Section F.10); under symbolic ambiguity BEARS and NeSyDM separate along an argmax-vs-calibration axis.

Limitations. Three assumptions bound the present results. (i) *Scale.* Our experiments use $|\Delta| = 2$; the compiled SDD grows empirically as $\approx |\Delta|^{2.9}$ with a $40\times$ compile-time jump at $|\Delta| = 4$ (Section D.5). Saturation and grounding are polynomial and the compilation is exact, but we do not demonstrate ontology-scale ABoxes such as SNOMED CT or the Gene Ontology; that regime requires lifted WMC and is the principal open problem. (ii) *Finite named domain.* MOOSE learns over a given finite ABox of named individuals (Definition 1); this defines the ABox-supervised task rather than weakening it, but it does not by itself perform open-domain inference over unnamed individuals. (iii) *Role-pinning.* Soundness of the Links encoding treats an existential as universal on the named pair, assuming role atoms over Sig_R^o are pinned by evidence (§3.2, Stage 7), a MOOSE-specific choice for the finite-named-domain regime, not an ELK theorem. The theoretical contribution, a Lean-verified exact compilation and the RS analysis, stands independently of the scaling outcome: it is the learned-perception pipeline at ontology scale, not the correctness result, that the scalability question concerns.

Open directions include lifted WMC [37] for larger ABox domains, $\mathcal{ALC}$ rewriting that would extend the same encoding to existential-on-the-left axioms, and evaluation on biomedical ontologies such as SNOMED CT and the Gene Ontology, and link prediction over latent role assertions: the role atoms $R(a, b)$ that the weight map (Table 1) currently fixes at $\frac{1}{2}$ when unobserved would instead be predicted from perception, extending the same WMC layer from latent-concept learning to latent-role learning.

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Supplemental Material Statement: Source code, $\mathcal{O}_{\text{MNIST}}$, the Pizzaiolo subset, seeds, per-seed JSON, the Lean 4 proof library, and pseudocode are at <https://github.com/bio-ontology-research-group/moose-iswc/>; full proofs are given in Section D.

Declaration of use of Generative AI. A large language model assisted with copy-editing and with Lean 4 transcription of the proofs in Section D. All technical content (definitions, theorem statements, proof structure, algorithms, experimental design and results) was authored, verified, and approved by the authors, and every mechanized statement was checked by the Lean type checker.

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A $\mathcal{EL}^{++}$ syntax, semantics, and ELK calculus

This appendix collects the formal foundations on which MOOSE is built: the syntax and semantics of the $\mathcal{EL}^{++}$ profile of OWL 2 EL, the Baader–Brandt–Lutz (BBL) normal-form rules NF1–NF7 [5], and the ELK saturation calculus [26] that drives Stage 1 of the compilation pipeline. Every construct, normal form, and rule listed here is also formalized in the Lean 4 library introduced below; Section D contains the full proofs of §4.

The Lean library. All formalized statements cited in this paper live in the Lean 4 library ELKSDD, distributed with the supplementary release at <https://github.com/bio-ontology-research-group/moose-iswc/>. Theorems are organized into namespaces: ELKSDD.ELpp (the bulk of the OWL 2 EL stack (syntax, semantics, the ELK saturation calculus, canonical-model completeness, the SCC compositional results, the DISPONTE correspondence, and the paper-citation theorems), ELKSDD.SDD (sentential decision diagrams and weighted model counting), ELKSDD.RangeNorm (the syntactic range-elimination of [6]), and the ELKSDD.EL and ELKSDD.MinIEL fragments for testing purposes. The DeepProbLog equivalence layer lives in a companion library under namespace Moose. Inline references of the form `<namespace>.<theorem>` point to a specific theorem; the index in Section D.10 cross-references every paper claim with its mechanized counterpart. Every theorem listed in Section D.10 is *audit-clean*: its proof depends only on the standard Lean foundation `{propext, Classical.choice, Quot.sound}` and the formalization contains zero `sorry` or `admit`. An audit pass in each library runs `#print axioms` on every paper-cited theorem at build time and reports its dependency set.

Syntax. Fix countably infinite, pairwise-disjoint vocabularies of *atomic concept names* N_C , *role names* N_R , and *individual names* N_I . $\mathcal{EL}^{++}$ *concepts* are built by the grammar

$$C, D ::= \top \mid \perp \mid A \mid \{a\} \mid C \sqcap D \mid \exists R.C,$$

where $A \in N_C$, $a \in N_I$, and $R \in N_R$. An *ontology* $\mathcal{O}$ is a finite set of axioms of one of the following shapes (Table 4): general concept inclusions $C \sqsubseteq D$, role inclusions $R \sqsubseteq S$, role chains $R_1 \circ R_2 \sqsubseteq S$, range restrictions $\text{range}(R) \sqsubseteq C$, reflexive-role declarations $\text{Ref}(R)$, local-reflexivity $\exists R.\text{Self}$, and key axioms $\text{HasKey}(C, R_1, \dots, R_n)$. The *ground vocabulary* of $\mathcal{O}$ over a finite individual set Δ is $\mathcal{V}_{\mathcal{O}}^{\Delta} := \{A(a) : A \in \text{Sig}_C(\mathcal{O}), a \in \Delta\} \cup \{R(a, b) : R \in \text{Sig}_R(\mathcal{O}), a, b \in \Delta\}$.

Semantics. An *interpretation* $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ comprises a non-empty domain $\Delta^{\mathcal{I}}$ and a mapping $\cdot^{\mathcal{I}}$ that sends each $A \in N_C$ to $A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$, each $R \in N_R$ to $R^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$, and each $a \in N_I$ to $a^{\mathcal{I}} \in \Delta^{\mathcal{I}}$. Concept and axiom semantics extend $\cdot^{\mathcal{I}}$ to compound concepts and constraints as in Table 4. $\mathcal{I}$ is a *model* of $\mathcal{O}$, written $\mathcal{I} \models \mathcal{O}$, if it satisfies every axiom in $\mathcal{O}$; $\mathcal{O} \models C \sqsubseteq D$ means $C^{\mathcal{I}} \subseteq D^{\mathcal{I}}$ in every model $\mathcal{I}$ of $\mathcal{O}$.

Table 4. $\mathcal{EL}^{++}$ syntax and semantics. Together with role inclusion, role chain, and range axioms, plus the ABox-style constructs (nominals, **Self**, **HasKey**), this is the OWL 2 EL profile minus datatype properties.

Construct	Syntax	Semantics
top	$\top$	$\Delta^{\mathcal{I}}$
bottom	$\perp$	$\emptyset$
atomic concept	A	$A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$
nominal	$\{a\}$	$\{a^{\mathcal{I}}\}$
conjunction	$C \sqcap D$	$C^{\mathcal{I}} \cap D^{\mathcal{I}}$
existential	$\exists R.C$	$\{a \mid \exists b : (a, b) \in R^{\mathcal{I}}, b \in C^{\mathcal{I}}\}$
local reflexivity	$\exists R.\mathbf{Self}$	$\{a \mid (a, a) \in R^{\mathcal{I}}\}$
GCI	$C \sqsubseteq D$	$C^{\mathcal{I}} \subseteq D^{\mathcal{I}}$
role inclusion	$R \sqsubseteq S$	$R^{\mathcal{I}} \subseteq S^{\mathcal{I}}$
role chain	$R_1 \circ R_2 \sqsubseteq S$	$R_1^{\mathcal{I}} \circ R_2^{\mathcal{I}} \subseteq S^{\mathcal{I}}$
range restriction	$\text{range}(R) \sqsubseteq C$	$\forall (a, b) \in R^{\mathcal{I}} : b \in C^{\mathcal{I}}$
reflexive role	$\text{Ref}(R)$	$(a, a) \in R^{\mathcal{I}} \text{ for all } a \in \Delta^{\mathcal{I}}$
key	$\text{HasKey}(C, R_1, \dots, R_n)$	shared R_i -fillers force individual identity

BBL normal forms NF1–NF7. Baader, Brandt, and Lutz [5] show every $\mathcal{EL}^{++}$ ontology can be transformed into one whose GCIs have one of seven shapes by introducing fresh atomic names, with the transformation conservative for atomic subsumptions: NF1, $A_1 \sqcap \dots \sqcap A_n \sqsubseteq B$ (atomic conjunctive LHS); NF2, $\exists R.A \sqsubseteq B$ (atomic existential LHS); NF3, $A \sqsubseteq \exists R.B$ (atomic existential RHS); NF4, $A \sqsubseteq \perp$ (atomic unsat); NF5, $A \sqsubseteq B$ (atomic-atomic); NF6, $\top \sqsubseteq A$ (top-LHS); NF7, $\perp \sqsubseteq A$ (vacuous). Our Lean formalization (namespace `ELKSDD.Normalize`) covers all seven rules via the canonical-extension construction of BBL 2005, §3.1.

ELK saturation calculus. Kazakov, Krötzsch, and Simančík [26] give a saturation procedure that decides Sat-derivability $\text{Sat}(\mathcal{O}, C, D)$ for the full OWL 2 EL profile (minus datatypes) in time $O(|\mathcal{O}|^4)$. The calculus operates on a finite collection of *atomic* subsumptions and *links* $E \xrightarrow{R} D$, closed under a fixed rule set: R_0 ($A \sqsubseteq A$), $R_{\top}$ ($A \sqsubseteq \top$), $R_{\sqsubseteq}$ (transitivity along axioms), $R_{\sqcap}^-$ and $R_{\sqcap}^+$ (conjunction), $R_{\perp}$ (unsatisfiability propagation), $R_{\exists}^+$ and $R_{\perp, \exists}$ (existential constructors), role-side rules R_{rinc} , R_{rchain} for role inclusions and chains, plus specialized rules for ranges, reflexive roles, **Self**, the nominal-handling rules of [26, §3], and the merging canonical-model construction of [26, §6] for shapes 1–4 of nominals together with **HasKey**. Saturation terminates because all derivable atoms are over the finite vocabulary of $\text{Sub}(\mathcal{O}) \cup \text{Roles}(\mathcal{O})$; the resulting closure has size $O(n^2 r)$ for $n = |\text{Sub}(\mathcal{O})|$ and $r = |\text{Roles}(\mathcal{O})|$ [26, Thm. 1]. Our formalization (namespace `ELKSDD.ELpp`) mechanizes the same calculus, sound and complete on the nominal-free fragment (with ranges, reflexive roles, **Self**, role inclusions, and role chains) and on the LHS-nominal and shallow-exist-RHS-nominal extensions; shapes 2/4 (RHS-nominal GCIs $C \sqsubseteq \{a\}$, $\{a\} \sqsubseteq \{b\}$) and **HasKey** are handled by the merging canonical-model construction of [26, §6], fully mechanized as `ELKSDD.ELpp.complete_via_mergedCanonRegular`

and `ELKSDD.ELpp.complete_via_mergedCanon_nom` (both audit-clean), under the shallow concept restriction of [26, §6]: concepts may not contain deep existentials with non-nominal targets. The unified completeness theorem `ELKSDD.ELpp.complete_owl2el` cases on a fragment witness to invoke the appropriate canonical-model construction; together with `ELKSDD.ELpp.sound_owl2el`, it gives $\text{Sat} \iff \text{Entails}$ on every ontology meeting at least one of its five branch preconditions. Every individual $\mathcal{EL}^{++}$ construct (nominals, ranges, role chains, reflexive, Self, HasKey) is supported in some branch, but the branches are not jointly compatible: shapes 2/4 (RHS-nominal GCIs $C \sqsubseteq \{a\}$, $\{a\} \sqsubseteq \{b\}$) combined with deep non-nominal existentials, and the merging fragment combined with role chains or ranges, hit the case-disabling preconditions of [26, §6] (an ontology may have a model where C is empty and another where C is forced into the $\{a\}$ -class; a single canonical construction must pick one); the supported shapes follow the consequence-based reasoning literature [48, 2]. The concurrent verified-reasoner effort VEL [42] mechanizes BBL 2005 $\mathcal{EL}^{++}$ in Coq but covers strictly less than our unified completeness theorem (no nominals, no RHS-nominal GCIs, no ranges, no role chains, no HasKey). The exact closure-size bound $|L| \leq 5n^3(r+1)$ that includes ranges, reflexive, and Self is `ELKSDD.ELpp.sat_closure_total_polynomial_bound`.

B Task formalism and MNIST instantiation

Definition 1 (ABox-supervised latent concept learning). *A training instance is a triple $(\Delta, \mathbf{x}, \mathbf{e})$ where $\mathbf{x} = (x_a)_{a \in \Delta}$ with each $x_a \in \mathcal{X}$ and $\mathbf{e}$ is a set of literals (truth-tagged ground atoms) over the observable ground vocabulary; ground atoms not in $\mathbf{e}$ are unobserved. The task is to learn a weight-shared per-individual classifier $f_\theta : \mathcal{X} \rightarrow [0, 1]^{\text{Sig}_C^\ell}$ such that $f_\theta(x_a)[C]$ approximates the marginal probability of $C(a)$ given $\mathcal{O}$ and $\mathbf{e}$, for every $C \in \text{Sig}_C^\ell$ and $a \in \Delta$. A negative literal $\neg C(a) \in \mathbf{e}$ encodes the polarity of a ground variable in $\mathcal{V}_\mathcal{O}^\Delta$ ($\mathcal{EL}^{++}$ itself has no concept negation); role atoms are not classified by f_θ , and latent role atoms are marginalized at the uniform prior.*

A boolean assignment $M : \mathcal{V}_\mathcal{O}^\Delta \rightarrow \{0, 1\}$ induces an ABox interpretation $\mathcal{I}_M = (\Delta, \cdot^M)$ via $C^M := \{a : M(C(a)) = 1\}$ and $R^M := \{(a, b) : M(R(a, b)) = 1\}$. The differentiable WMC layer evaluates $\text{WMC}(\alpha; w, \mathbf{e})$ where the literal-weight map w is the unique extension of $f_\theta(\mathbf{x})$ and $\mathbf{e}$ that makes the training-loss, posterior, and entailment-query operations of §3.4 three settings of the same SDD traversal:

The literal-weight map itself is now inlined in §3.1 as Table 1.

Example 1 (MNIST instantiation). Definition 1 instantiates on MNIST as follows. The TBox is $\mathcal{O}_{\text{MNIST}}$ over $\text{Sig}_C = \{D_0, \dots, D_9, \text{Even}, \text{Odd}, \text{Prime}, \text{Composite}\}$ and $\text{Sig}_R = \{\text{succ}, \text{plus_two}\}$ (full listing in Section F.2). The observable signature is $\text{Sig}_C^\circ = \{\text{Even}, \text{Odd}, \text{Prime}, \text{Composite}\}$; the latent signature is $\text{Sig}_C^\ell = \{D_0, \dots, D_9\}$ (the digit class is never revealed). A training instance places one MNIST image x_a at each named individual and supplies one or two property literals as evidence, e.g. $\mathbf{e} = \{\text{Even}(a), \neg \text{Prime}(a)\}$ for an image of digit 4. An

Table 5. Assumptions introduced by each component of the MOOSE pipeline and their departure from open-world semantics.

Component	Assumption	Departure from open-world semantics
EL fragment	Background axioms lie in the $\mathcal{EL}^{++}$ profile: conjunction and existentials, no right-hand-side disjunction or cardinality.	Exhaustiveness and counting are not natively expressible; they must be supplied as closure axioms rather than read from the ontology.
Closed domain	Learning and inference range over a given finite set of named individuals.	No certain answers are produced over anonymous or unnamed individuals; entailment over the anonymous domain is not performed.
Role-pinning	An existential $\exists R.C$ is scored on the observed role pair, with role atoms over Sig_R^o fixed by the evidence.	Unobserved role witnesses are not hypothesized; existential import over the anonymous domain is dropped.
Declared families	Mutual exclusion and covering hold only on families the model declares exhaustive.	The closed-world reading is logical and opt-in; outside declared families the open-world reading is retained.
Factorized perception	The predictor factorizes $p_\theta(\mathbf{c} \mid \mathbf{x}) = \prod_i p_\theta(c_i \mid \mathbf{x})$: atoms independent given the image.	Joint perceptual correlations among atoms are not modeled; this is the reasoning-shortcut source, mitigated but not removed by BEARS and NeSyDM.

independent per-atom baseline matching its predictions to the observed literals provably cannot disambiguate the latent atoms when several digits share the property profile (e.g. D_3, D_5, D_7 are all $\text{Odd} \sqcap \text{Prime}$) [39,41,29].

B.1 Summary of pipeline assumptions

Table 5 collects the assumptions that the pipeline introduces beyond the source ontology, one per component, and states for each what it concedes relative to the open-world semantics that OWL EL is designed for. The assumptions are not defects of the implementation but scoping choices that define the ABox-supervised learning task (Definition 1); the right-hand column makes explicit what a user gives up by adopting them.

C Compilation algorithm and worked example

This appendix gives the seven-stage compilation pseudocode (Algorithm 1) and works a concrete two-axiom ontology through it (Section C.3).

C.1 Pseudocode

Algorithm 1 gives the full pseudocode of the seven-stage compilation algorithm summarized in §3.2. Stages 2–5 (Clark completion of the saturation output, SCC detection via Tarjan’s algorithm, SCC-local time-stamped unrolling, and Tseitin CNF conversion) operate on the TBox-level derived atoms and produce a propositional CNF that exactly characterizes the ELK canonical model. For acyclic ontologies (all SCCs are trivial), Stages 3–4 are no-ops and the Clark biconditionals compile directly to CNF. Two implementation choices are essential for the soundness, completeness, and termination arguments. The first is the *signature side condition* on the existential-introduction rule $R_{\exists}^+$: the rule fires only on existentials already in $\text{Sub}(\mathcal{O})$ (line 1 of the algorithm). This is the standard ELK rule [26, §5] and is what keeps saturation finite on ontologies with cyclic role inclusions. The second is *uniform super-role materialization* under the role-composition rule $R_{\circ}$: the implementation routes every link insertion through a helper that records the link for the base role and, recursively, for every strict super-role. Without this step, an ontology with $R_1 \circ R_2 \sqsubseteq S$ and $S \sqsubseteq T$ produces the S -link from the chain rule but silently loses the T -link, breaking ELK completeness (Theorem 5) on chain heads with a super-role.

C.2 Stages 2–5: the TBox-level encoding

This subsection expands the Stage 2–5 block of Algorithm 1, which turns the Stage-1 saturation into a propositional TBox theory. For NeSy learning these stages are diagnostic (§3.2); they are the compilation path used for probabilistic ontology reasoning, where the input-axiom variables carry weights.

Stage 2: Clark completion. The saturation of Stage 1 can be viewed as a monotone Datalog program [18]: each derived atom h has one or more grounded rule body $body_1, \dots, body_k$ that can derive it. The *Clark completion* replaces these defining clauses with a biconditional

$$h \longleftrightarrow body_1 \vee body_2 \vee \dots \vee body_k, \quad (4)$$

asserting that h holds if and only if at least one of its justifications holds. Each $body_j$ is a conjunction of previously derived atoms and *input axiom variables* α_i representing ontology axioms. The input variables are free (not defined by any biconditional) and represent whether each axiom is active: in standard reasoning all α_i are true; in probabilistic reasoning each carries a weight. When the dependency graph is acyclic, Clark completion has the Datalog least fixed point as its only model for each assignment to the input variables [18]. For recursive monotone Datalog, however, completion alone can also admit supported non-least fixed points. Stages 3–4 therefore identify recursive strongly connected components and replace their circular definitions by time-stamped unrolling from the all-false interpretation. The completed and unrolled theory, rather than Clark completion alone, has exactly one model; with all α_i true, that model is exactly the set of atoms ELK derives from $\mathcal{O}$, i.e. the saturation closure $\text{Sat}(\mathcal{O})$ [26].

Algorithm 1 MOOSE: $\mathcal{EL}^{++}$ ontology to SDD over ground atoms

Require: Ontology $\mathcal{O}$ as a set of GCIs $C \sqsubseteq D$, role inclusions $R \sqsubseteq S$, role chains $R_1 \circ R_2 \sqsubseteq S$. Finite ABox domain Δ .

Ensure: SDD α over $\mathcal{V}_{\mathcal{O}}^{\Delta}$ (Theorem 1).

[Stage 1: ELK saturation]

- 1: $\mathcal{E}_{\exists} \leftarrow \{\exists R.D \mid \exists R.D \in \text{Sub}(\mathcal{O})\}$ ▷ signature side condition on $R_{\exists}^+$
- 2: $(\Sigma, \Lambda) \leftarrow \text{Saturate}(\mathcal{O}, \mathcal{E}_{\exists})$ ▷ Σ : subsumptions $C \sqsubseteq D$; Λ : links $E \xrightarrow{R} C$

[Stages 2–5: Clark completion, SCC, unrolling, CNF]

- 3: **for** each derived atom $h \in \Sigma \cup \Lambda$ **do**
- 4: $\text{Clark}(h) \leftarrow h \leftrightarrow \bigvee_j (\bigwedge_{b \in \text{body}_j(h)} b)$ ▷ Stage 2: Clark completion
- 5: **end for**
- 6: $\text{SCCs} \leftarrow \text{Tarjan}(\text{DepGraph}(\text{Clark}))$ ▷ Stage 3: SCC detection
- 7: **for** each non-trivial SCC S of size $T=|S|$ ($|S| > 1$ or self-loop) **do** ▷ Stage 4:
- time-stamped unrolling
- 8: **for** each $h \in S$ **do**
- 9: emit $x_h^{(0)} \leftrightarrow \perp$; $x_h \leftrightarrow x_h^{(T)}$
- 10: **end for**
- 11: **for** each $h \in S$, $t = 0, \dots, T-1$ **do**
- 12: emit $x_h^{(t+1)} \leftrightarrow x_h^{(t)} \vee \bigvee_j (\bigwedge_b x_b^{(t)})$
- 13: **end for**
- 14: **end for**
- 15: $\text{CNF} \leftarrow \text{Tseitin}(\text{all formulas})$ ▷ Stage 5: CNF conversion

[Stages 6–7: ABox grounding and SDD compilation]

- 16: $\text{AtomSub} \leftarrow \{(X, Y) \in \Sigma : X, Y \in \text{Sig}_C(\mathcal{O}), X \neq Y\}$
- 17: $\text{Disj} \leftarrow \{\text{flatten}(C) : (C \sqsubseteq \perp) \in \Sigma, C \text{ atomic conjunction}\}$
- 18: $\text{Unsat} \leftarrow \{X \in \Sigma : X \sqsubseteq \perp, X \in \text{Sig}_C(\mathcal{O})\}$
- 19: $\text{Links} \leftarrow \{(X, R, Y) \in \Lambda : X, Y \in \text{Sig}_C(\mathcal{O})\}$
- 20: $\Gamma \leftarrow \emptyset$
- 21: **for** $(X, Y) \in \text{AtomSub}$, $a \in \Delta$ **do** $\Gamma \cup = \{\neg X(a) \vee Y(a)\}$
- 22: **end for**
- 23: **for** $(X_1, \dots, X_n) \in \text{Disj}$, $a \in \Delta$ **do** $\Gamma \cup = \{\bigvee_i \neg X_i(a)\}$
- 24: **end for**
- 25: **for** $X \in \text{Unsat}$, $a \in \Delta$ **do** $\Gamma \cup = \{\neg X(a)\}$
- 26: **end for**
- 27: **for** $(X, R, Y) \in \text{Links}$, $(a, b) \in \Delta^2$, $a \neq b$ **do**
- 28: $\Gamma \cup = \{\neg X(a) \vee \neg R(a, b) \vee Y(b)\}$ ▷ NF3 forward
- 29: **if** $|\{Y' : (X, R, Y') \in \text{Links}\}| = 1$ **then**
- 30: $\Gamma \cup = \{\neg R(a, b) \vee \neg Y(b) \vee X(a)\}$ ▷ NF4 reverse
- 31: **end if**
- 32: **end for**
- 33: **return** $\text{SDDCompile}(\Gamma)$ ▷ PySDD bottom-up

When the ontology contains axioms with existential restrictions, derived atoms can have heads that contain nested existentials, e.g. $x_{D_i} \sqsubseteq \exists R. \exists R. D_k$. For NeSy training, the Clark completion is restricted to atoms whose head is an atomic concept, a conjunction of atomic concepts, $\top$, or $\perp$ (the *groundable* atoms); all other atoms are filtered out. This keeps Stages 3–5 tractable without losing any ABox-relevant consequence, because the filtered atoms have no corresponding ground variable in $\mathcal{V}_O^A$.

Stage 3: SCC detection. The dependency graph among derived atoms may contain cycles. For example, a cyclic successor chain $D_0 \sqsubseteq \exists \text{succ}. D_1 \sqsubseteq \exists \text{succ}. \exists \text{succ}. D_2 \sqsubseteq \dots \sqsubseteq D_0$ (through the existential derivations) produces circular dependencies among the derived concept-level atoms. Plain biconditionals have circular definitions in this case. We apply Tarjan’s algorithm [47] to find strongly connected components (SCCs) of the dependency graph. Acyclic atoms (trivial SCCs) can use the plain biconditional of Equation (4) directly. Non-trivial SCCs (size > 1 , or self-loops) require time-stamped unrolling in Stage 4.

Stage 4: propositional compilation with time-stamped unrolling. For acyclic atoms the biconditional of Equation (4) is directly expressible in propositional logic. For cyclic SCCs we break the circularity by introducing time-stamped copies $x_h^{(0)}, x_h^{(1)}, \dots, x_h^{(T)}$ of each atom h within the SCC and defining:

$$x_h^{(0)} \longleftrightarrow \perp, \quad (5)$$

$$x_h^{(t+1)} \longleftrightarrow x_h^{(t)} \vee \bigvee_j \left(\bigwedge_{b \in \text{body}_j(h)} x_b^{(t)} \right), \quad (6)$$

$$x_h \longleftrightarrow x_h^{(T)}, \quad (7)$$

where the local bound $T = |S|$ (the SCC size) suffices because each iteration of the monotone operator must derive at least one new atom or the fixed point is already reached. Only atoms within the *same* non-trivial SCC receive time-stamped copies; atoms from other SCCs (already computed in topological order) use their plain variable. This gives total cost $\sum_S |S|^2$ instead of $|\mathcal{V}|^2$ for global unrolling.

Stage 5: CNF conversion. The Tseitin transformation [50] converts the nested propositional formulas from Stage 4 into equisatisfiable conjunctive normal form (CNF) by introducing an auxiliary variable for each sub-formula φ and adding clauses that enforce the auxiliary to equal φ . The resulting CNF is linear in the size of the Stage 4 output and encodes the full TBox-level propositional theory. In the probabilistic ontology reasoning use case, this CNF is the input to an SDD compiler that produces a circuit over the input-axiom variables; in the NeSy learning use case, the CNF serves a diagnostic role (characterizing the TBox complexity) and the ABox grounding of Stage 6 produces its own clause set over $\mathcal{V}_O^A$.

WMC recursion. The differentiable WMC layer of Stage 7 traverses the compiled SDD α and evaluates

$$\text{WMC}(\alpha; w) = \begin{cases} w(\ell) & \text{if } \alpha = \ell \text{ is a literal node,} \\ 0 & \text{if } \alpha = \perp, \\ 1 & \text{if } \alpha = \top, \\ \sum_{(p_i, s_i) \in \alpha} \text{WMC}(p_i; w) \cdot \text{WMC}(s_i; w) & \text{if } \alpha \text{ is a decision node,} \end{cases} \quad (8)$$

with literal weights w read from Table 1.

C.3 Worked example

Consider the two-axiom fragment $\mathcal{O}_5 := \{D_5 \sqsubseteq \text{Odd}, D_5 \sqsubseteq \text{Prime}\}$ on $\Delta = \{a\}$, drawn from $\mathcal{O}_{\text{MNIST}}$ (§5.1). **Stage 1** runs the ELK rule $R_{\sqsubseteq}$ over $\mathcal{O}_5$ and adds the two input axioms to Σ as atomic subsumptions; no role axioms, so $\Lambda = \emptyset$. Stages 2–5 (Clark completion, SCC detection, propositional compilation, CNF conversion) produce a propositional theory over the derived atoms; for this acyclic two-axiom fragment no time-stamped unrolling is needed (both atoms are in trivial SCCs). **Stage 6** applies the four extractors to (Σ, Λ) : both subsumptions match the **AtomSub** shape (atomic-atomic, $X \neq Y$), so $\text{AtomSub} = \{(D_5, \text{Odd}), (D_5, \text{Prime})\}$; the other extractors produce nothing. The extractors ground each $(X, Y) \in \text{AtomSub}$ on every $a \in \Delta$, emitting $\Gamma = \{\neg D_5(a) \vee \text{Odd}(a), \neg D_5(a) \vee \text{Prime}(a)\}$ over $\mathcal{V}_{\mathcal{O}_5}^{\{a\}}$. **Stage 7** compiles Γ into an SDD whose models are exactly the assignments in which $D_5(a) = 1$ implies $\text{Odd}(a) = 1$ and $\text{Prime}(a) = 1$. The full $\mathcal{O}_{\text{MNIST}}$ extends $\mathcal{O}_5$ with the analogous property axioms for the remaining digits, the 45 pairwise digit disjointness axioms (handled by **Disj**), the 10 NF3 successor existentials, and the NF7 role chain (Section F.2); the SDD is 155 nodes over 14 atoms and compiles in under a quarter of a second (Section F.1).

D Full proofs

This part of the appendix contains the full proofs of the §4 theorems (Theorems 1 to 4), the ELK mechanization of the reused infrastructure (Theorems 5 and 6, Section D.1), and the closure-augmented and inference-time variants (Theorems 7 and 8). The final subsection (Section D.10) cross-references each paper claim with its Lean module and theorem identifier. We use the notation introduced in §4.

Mechanized statements. The Lean library **ELKSDD** (Section A) proves every prior-work result it relies on rather than admitting it as an axiom: ELK soundness/completeness [26] is proved end-to-end in namespaces **ELKSDD.EL** and **ELKSDD.ELpp**; SDD compilation correctness [21,17] and the WMC-correctness theorem underlying Theorem 8(c) are proved structurally in namespace **ELKSDD.SDD**;

the four ELK extractors are concrete `noncomputable defs` with proven membership and size theorems; the Manhaeve 2018 distribution-semantics step of the DPL-equivalence proof is closed both qualitatively (`Iff.rfl`) and quantitatively ([21] SDD-WMC correctness, by structural induction). Running `#print axioms` on every paper-cited theorem at build time confirms that the dependency set is exactly `{propext, Classical.choice, Quot.sound}`.

D.1 Reused ELK infrastructure: soundness, completeness, and polynomial Sat decision

The two results below are restatements of the standard ELK soundness/completeness theorem and the polynomial-time Sat decision procedure of Kazakov, Krötzsch, and Simančík [26]. They are reused unchanged in §4 but re-mechanized in our Lean library so that Theorems 1 to 4 can be proved within the same audit-clean foundation. `OWL2ELFragment` abbreviates the syntactic fragment witness selecting the canonical-model construction used for completeness on $(\mathcal{O}, C, D)$: (a) nominal-free with C, D in the signature of $\mathcal{O}$ and range-chain-safe; (b) the ABox-style LHS-nominal extension; (c) the shallow-exist RHS-nominal extension; (d) the strict fragment without ranges or chains; (e) the merging canonical model [26, §6] for shapes 1–4 of nominals and `HasKey`, restricted to Shallow concepts (no deep existentials with non-nominal targets). The disjunction is the inductive predicate `ELKSDD.ELpp.OWL2ELFragment`; together with `OWL2ELFragment` the unified completeness theorem `ELKSDD.ELpp.complete_owl2el` covers any ontology meeting at least one branch precondition (a)–(e). Every individual $\mathcal{EL}^{++}$ construct is supported in some branch; the residual unmechanized cases (shapes 2/4 (RHS-nominal GCIs) combined with deep non-nominal existentials, and the merging fragment combined with ranges or chains) correspond to the case-disabling preconditions of [26, §6] and constitute open work in the ELK literature itself rather than gaps in the Lean formalization (see Section A for discussion of a 2019 survey, a 2025 proof-theoretic study, and the concurrent VEL effort, none of which extend the canonical-model coverage). The atomic-shape vocabulary $\mathcal{V}_{\mathcal{O}}^A$ used by the SDD encoding (§3.2) is a separate, deliberate scope choice: nested existentials produce ELK-internal derivations but have no ground counterpart in the SDD; ELK Sat-completeness via the unified completeness theorem holds over all concepts, the SDD-encoding completeness holds over the atomic-shape ABox slice (Theorem 1).

Theorem 5 (ELK soundness and completeness on $\mathcal{EL}^{++}$). *For every $\mathcal{EL}^{++}$ ontology $\mathcal{O}$ and concepts C, D , if `OWL2ELFragment`($\mathcal{O}, C, D$) holds then*

$$\text{Sat}(\mathcal{O}, C, D) \iff \mathcal{O} \models C \sqsubseteq D. \quad (9)$$

Proof. Soundness (left to right) is unconditional in `OWL2ELFragment` and proved by structural induction on the ELK derivation, mechanized as `ELKSDD.ELpp.sound_owl2el`. Completeness (right to left) is by case-split on

OWL2ELFragment to one of five canonical-model constructions (Kazakov 2014 §3, §6 for the merging case); mechanized as `ELKSDD.ELpp.complete_owl2e1`. See [26] for the original arguments. $\square$

Theorem 6 (Polynomial Sat decision). *For every $\mathcal{EL}^{++}$ ontology $\mathcal{O}$ there exists a list L of pairs of concepts with $|L| \leq |\text{Sub}(\mathcal{O})|^2$ such that for every $C, D \in \text{Sub}(\mathcal{O})$,*

$$\text{Sat}(\mathcal{O}, C, D) \iff (C, D) \in L.$$

Proof. The witness L is the time-stamped Clark closure `derivableClosure`($\mathcal{O}$): length bound by `derivableClosure.length` (counts concept pairs in the saturation closure); membership equivalence by `sat_iff_in_derivableClosure` (simultaneous induction over the Sat constructors). Mechanized as `ELKSDD.ELpp.sat_decision_polynomial`; an exact closure-size bound $|L| \leq 5n^3(r+1)$ that includes ranges, reflexive, and Self ($n = |\text{Sub}(\mathcal{O})|$, $r = |\text{Roles}(\mathcal{O})|$) is `ELKSDD.ELpp.sat_closure_total_polynomial_bound`, with per-shape exact lengths. See [26, Thm. 1, §5] for the original $O(|\mathcal{O}|^4)$ saturation runtime. $\square$

D.2 Verified SDD encoding (Theorem 1)

Proof (Proof of Theorem 1). The witness is $t = \text{cmp}(\mathcal{O}, C, D)$, mechanized in namespace `ELKSDD.ELpp` (with the SDD-side WMC theorems in namespace `ELKSDD.SDD`). Conjunct (1): the compiled Shannon tree’s models are exactly the worlds whose selected sub-ontology is Sat-derivable. Conjunct (2): a Shannon-recursive unfolding of WMC enumerating the $2^{|\mathcal{O}|}$ leaves. Conjunct (3): by induction on $|\mathcal{O}|$, a Shannon node at depth k has $2^{|\mathcal{O}|-k+1} - 1$ descendants. Lean theorem identifiers are listed in Table 6. $\square$

The standard probabilistic-DL “entailment-as-WMC-zero” phrasing [44] follows immediately: at uniform per-axiom prior $w \equiv \frac{1}{2}$, $\text{WMC}^{\mathbb{Q}} = 0$ iff no world’s selected sub-ontology entails $C \sqsubseteq D$ (Theorem 2 combined with Theorem 5). The production four-extractor ground encoding (§3.2, Stages 6–7) is a separate ABox encoding whose soundness over the finite domain follows from Theorem 5 (Lean: `ELKSDD.ELpp.sound_owl2e1` and `ELKSDD.ELpp.complete_owl2e1`) under the partial-supervision role-pinning of §3.2.

D.3 Correctness of the closure-augmented circuit (Theorem 7)

Theorem 7 (Correctness of the closure-augmented circuit). *For every modeller-declared exhaustive family $\mathcal{F} = \{D_0, \dots, D_{K-1}\}$, define $\text{match}_{\pi}(a)$ as the conjunction of the positive and negated property literals specified by profile π . Then $M \models \alpha_{\text{clos}}$ iff $M \models \alpha_{\text{EL}}$ and, at every $a \in \Delta$, both of the following hold: (i) the partition constraints $\{\neg(D_i(a) \wedge D_j(a)) : i < j\} \cup \{D_0(a) \vee \dots \vee D_{K-1}(a)\}$; and (ii), for every distinct declared profile π , the reverse implication $\text{match}_{\pi}(a) \rightarrow \bigvee_{i: \pi_i = \pi} D_i(a)$.*

Proof (Proof of Theorem 7). $\alpha_{\text{clos}} = \text{SDDCompile}(\Gamma_{\text{EL}} \cup \Phi_{\text{clos}})$ by construction. For each marked family and individual, Φ_{mutex} contains the $\binom{K}{2}$ clauses $\neg D_i(a) \vee \neg D_j(a)$, and Φ_{cover} contains $D_0(a) \vee \dots \vee D_{K-1}(a)$. These are exactly the partition constraints in clausal form. For every distinct profile π , Φ_{profile} additionally contains the clausal form of $\text{match}_{\pi}(a) \rightarrow \bigvee_{i: \pi_i = \pi} D_i(a)$. Therefore, satisfaction of the full clause set is equivalent to conditions (i)–(ii). Generic SDD compilation correctness then gives the stated model equivalence [17,21]. The Lean theorem `ELKSDD.SDD.compile_correct` quantifies over the complete input clause list, so it applies to all three components of Φ_{clos} . $\square$

D.4 Production-encoding size

Theorem 6 bounds the saturation closure by $|\text{Sat}(\mathcal{O})| \leq 5n^3(r+1)$ atoms with $n = |\text{Sub}(\mathcal{O})|$ and $r = |\text{Roles}(\mathcal{O})|$ (Lean: `ELKSDD.ELpp.sat_closure_total_polynomial_bound`). Grounding the extractors of §3.2 over $|\Delta| = m$ named individuals adds an at-most m^2 multiplicative factor (`AtomSub`, `Disj`, `Unsat` contribute m clauses each; `Links` contributes up to m^2 for the NF3/NF4 clauses). Each declared exhaustive family of size K adds a further $\binom{K}{2}m + (K+1)m$ closure clauses ($\binom{K}{2}m$ mutex, m covering, and up to Km profile-keyed reverse implications). SDD compilation of the resulting CNF is treated in Section D.5.

D.5 SDD treewidth dependence and empirical scaling

Bottom-up SDD compilation [17] of a CNF Γ produces an SDD whose worst-case size is exponential in $|\Gamma|$ [21]. For CNFs whose primal graph has treewidth w , the resulting SDD admits a polynomial bound $O(|\Gamma| \cdot 2^w)$ because tractable WMC over bounded-treewidth instances is reducible to compact circuit compilation. Empirically on MNIST the SDD node count fits $\propto |\Delta|^{2.9}$ for $|\Delta| \leq 3$ (consistent with the bounded-treewidth regime), with a $40\times$ compile-time jump between $|\Delta| = 3$ and $|\Delta| = 4$ as the primal-graph treewidth rises; at the $|\Delta| = 2$ configurations used in §5 compilation finishes well below one second on a CPU ($1\times$ RTX 4090, single CPU core; Section F.1).

D.6 Inference complexity (Theorem 8)

Theorem 8 (Inference complexity). *Let $|\alpha|$ denote the node count of either α_{EL} or α_{clos} . The training loss Equation (11), conditional posterior Equation (12), entailment query Equation (13), and one training step (loss plus gradient) all run in $O(|\alpha|)$ time [21,17], with the training step adding the cost of one forward and backward pass of f_{θ} .*

Proof (Proof of Theorem 8). The SDD α produced by Algorithm 1 is smooth, decomposable, and deterministic by PySDD construction [21,17]; let $|\alpha|$ denote its node count.

(a) WMC. The recursive traversal of §3.2 visits each node once (memoized by node id), contributing $p_{C(a)}$ or $1 - p_{C(a)}$ at literal nodes, 0 or 1 at constants, and

$\sum_i \text{WMC}(\text{prime}_i) \cdot \text{WMC}(\text{sub}_i)$ at decision nodes. Determinism makes each sum disjoint and decomposability makes each product over disjoint variable scopes, so $\text{WMC}(\alpha; p, \mathbf{e})$ is exact and runs in $O(|\alpha|)$ time. Each per-node operation is differentiable in the literal weights, and PyTorch autograd composes the backward pass through the same traversal in $O(|\alpha|)$ additional time.

(b) Conditional posterior. The ratio $P_\theta(C(a) \mid \mathbf{x}, \mathbf{e}) = \text{WMC}(\alpha; p_\theta(\mathbf{x}), \mathbf{e} \cup \{C(a) := \text{true}\}) / \text{WMC}(\alpha; p_\theta(\mathbf{x}), \mathbf{e})$ is two applications of (a), each $O(|\alpha|)$. The denominator is strictly positive whenever $\mathbf{e}$ is consistent with $\mathcal{O}$.

(c) Entailment query. Setting $p \equiv \frac{1}{2}$ gives the uniform prior; conditioning on $\mathbf{e} \cup \{C(a) := \text{false}\}$ clamps the corresponding literal. By Theorem 5 the SDD’s models are exactly the finite-domain interpretations satisfying the entailed clauses, so $\text{WMC}(\alpha; \frac{1}{2}, \mathbf{e} \cup \{C(a) := \text{false}\}) = 0$ iff no model of α has $C(a) = \text{false}$, iff $\mathcal{O} \cup \mathbf{e} \models C(a)$. The check is one WMC call, $O(|\alpha|)$.

(d) Training step. The loss $L(\theta; \mathbf{x}, \mathbf{e}) = -\log \text{WMC}(\alpha; p_\theta(\mathbf{x}), \mathbf{e})$ is one WMC call, $O(|\alpha|)$. Its gradient with respect to θ is computed by autograd through the WMC traversal (cost $O(|\alpha|)$) followed by the perception backbone’s backward pass (cost equal to the forward pass of f_θ). Total: $O(|\alpha|) + O(\text{cost of } f_\theta(\mathbf{x}))$. $\square$

D.7 Structural circuit properties

We define three structural properties used to characterize the tractability of weighted model counting on a circuit [21]. A *circuit* over variables $\mathbf{V}$ is a rooted directed acyclic graph whose leaves are literals over $\mathbf{V}$ (or the constants $\top$ / $\perp$), and whose internal nodes are either *product nodes* ($\wedge$, evaluating to the conjunction of their children) or *sum nodes* ($\vee$, evaluating to the disjunction of their children). Let φ be such a circuit; the *scope* of a node is the set of variables appearing in the subcircuit rooted at it, and the *support* is the set of total assignments to $\mathbf{V}$ on which the node evaluates to a non-zero value.

Definition 2 (Decomposability). *A circuit is decomposable if at every product node the scopes of its children are pairwise disjoint.*

Definition 3 (Determinism). *A circuit is deterministic if at every sum node the children have pairwise disjoint supports.*

Definition 4 (Smoothness). *A circuit is smooth if at every sum node all children share the same scope.*

A Sentential Decision Diagram (SDD) is a circuit whose internal nodes are *decision nodes* of the form $(p_1, s_1) \vee \cdots \vee (p_k, s_k)$, abbreviating $\bigvee_i (p_i \wedge s_i)$; each p_i (the *prime*) and s_i (the *sub*) are themselves SDDs over disjoint variable sets fixed by a v-tree, with the prime ranging over the variables of the left subtree and the sub over those of the right [20,17]. This prime/sub decomposition realizes all three structural properties above by construction: every prime p_i has scope disjoint from its corresponding sub s_i (decomposability), the primes are pairwise inconsistent (determinism), and they cover the input space allotted to the node by its v-tree (smoothness) [20,17]. Together, these properties make WMC linear in $|\alpha|$ and reduce conditioning, marginalization, and MAP inference to circuit traversals [21].

D.8 Compositional SCC factorization lemmas (Theorems 3 and 4)

The following lemmas establish the SCC factorization formalized in Lean: an arbitrary two-component partition of an ontology, $\mathcal{O} = \mathcal{O}_1 \cup \mathcal{O}_2$, with disjoint atom and role signatures, factors the closure exactly, modulo a single “global inconsistency” disjunct that captures the case where $\mathcal{O}_2$ alone derives $\top \sqsubseteq \perp$ (which then propagates to all of $\mathcal{O}_1 \cup \mathcal{O}_2$ via the $R_\perp$ rule, regardless of $\mathcal{O}_1$). This is what allows the algorithm to compile per-SCC SDDs and combine them, rather than compiling one monolithic SDD over the union.

Lemma 1 (Semantic SCC factorization). *Let O_1, O_2 be $\mathcal{EL}^{++}$ ontologies (finite sets of GCIs, role inclusions, and role chains). Let $\text{atoms}(O_i) \subseteq \text{Sig}_C$ and $\text{roles}(O_i) \subseteq \text{Sig}_R$ denote the sets of atomic concept names and role names appearing in axioms of O_i . Assume signature disjointness:*

$$\text{atoms}(O_1) \cap \text{atoms}(O_2) = \emptyset, \quad \text{roles}(O_1) \cap \text{roles}(O_2) = \emptyset. \quad (\text{Disj})$$

Then for every pair of $\mathcal{EL}^{++}$ concepts C, D whose atom and role names lie entirely in O_1 ’s signature ($\text{atoms}(C) \cup \text{atoms}(D) \subseteq \text{atoms}(O_1)$ and $\text{roles}(C) \cup \text{roles}(D) \subseteq \text{roles}(O_1)$), the closure factors as

$$\text{Sat}(O_1 \cup O_2) C \sqsubseteq D \iff \text{Sat}(O_1) C \sqsubseteq D \vee \text{Sat}(O_2) \top \sqsubseteq \perp. \quad (10)$$

The statement holds for the nominal-free, range-chain-safe fragment of full OWL 2 EL ($\mathcal{EL}^{++}$ minus concrete domains): both O_1 and O_2 are nominal-free, both satisfy the range-chain safety condition `ELKSDD.ELpp.RangeChainSafe` (range axioms are forbidden on the rinc-ancestors of any role-chain target, so that range-guards transfer vacuously through chain composition; [26] §3.3), and C, D are nominal-free. No restriction on $\top$ - or $\perp$ -axioms is imposed on either side.

OWL 2 EL nominals (ObjectOneOf singleton classes) are supported by the algorithm at the axiom level (normal forms `NFnomR` “ $A \sqsubseteq \{a\}$ ” and `NFnomL` “ $\{a\} \sqsubseteq B$ ”) and at the concept level (`Concept.nom`, `Interp.indiv`, `conceptIndividuals`, `ontologyIndividuals`) in our Lean implementation. The proof of Lemma 1 below covers the nominal-free fragment. Shape 1 nominals (ABox-style `ClassAssertion` axioms $\{a\} \sqsubseteq D$ with D nominal-free) are formalized by the relaxed `ELKSDD.ELpp.Sat_factor_nomLHS` theorem, under the `OntologyNomLHS` precondition (which permits LHS-nominal GCIs on the analyzed side O_1) and `AllNomInhabited` (every nominal index is consistent). The key observation is that an LHS-nominal $\{a\}$ pins the `prodInterp` evaluation to the specific point $\langle \mathcal{I}_1.\text{indiv}(a), b_0 \rangle$, so no general nominal-evaluation lemma is needed: the existing `P2` (which is already general in the second coordinate) is applied at $b = b_0$. Shapes 2/4 (RHS nominal: $A \sqsubseteq \{a\}$ or $\{a\} \sqsubseteq \{b\}$) and `HasKey` are handled by the merging canonical-model construction [26] (concept equivalence classes through nominals), fully mechanized in namespace `ELKSDD.ELpp` under the Shallow concept restriction of [26, §6]. The Sat-level SCC factorization theorem in this lemma is currently stated only for the

nominal-free and LHS-nominal cases; lifting the WMC factorization to the merge fragment is a straightforward composition of `ELKSDD.SCC.Sat_factor_refined` with `ELKSDD.ELpp.complete_via_mergedCanon_regular`.

Proof (Proof outline (full proof: `ELKSDD.SCC.Sat_factor_refined`)). The $(\Leftarrow)$ direction is monotonicity of `Sat` in the ontology, plus the $R_\perp$ rule when the second disjunct holds. For $(\Rightarrow)$ we may assume $\neg \text{Sat}(O_2) \top \sqsubseteq \perp$ (else the second disjunct is immediate), so the canonical (term) model $\mathcal{I}^{\text{can}}(O_2)$ of Theorem 5 is non-empty. Given an arbitrary $\mathcal{I}_1 \models O_1$ on Δ_1 , build the *product interpretation* $\mathcal{I}'$ on $\Delta_1 \times \mathcal{I}^{\text{can}}(O_2)$ that evaluates O_1 -signature atoms and roles on the first coordinate and O_2 -signature ones on the second. Signature disjointness (`Disj`) lets the evaluation of any O_i -signature concept factor cleanly through the corresponding component (Lean: `eval_prodInterp_01`, `eval_prodInterp_02`), which in turn gives $\mathcal{I}' \models O_1 \cup O_2$ (`prodInterp_satisfies`). By ELK soundness on the union, $\mathcal{I}' \models C \sqsubseteq D$; factoring back through the first coordinate at the basepoint $\langle a, x_\top \rangle$ yields $\mathcal{I}_1 \models C \sqsubseteq D$. Since $\mathcal{I}_1$ was arbitrary, ELK completeness on O_1 closes the left-hand disjunct. $\square$

Lemma 2 (Per-world Sat factorization). *Let $O = O_1 \cup O_2$ satisfy the signature-disjointness, nominal-free, and range-chain-safe preconditions of Lemma 1, and additionally assume O_2 consistent ($\neg \text{Sat}(O_2, \top, \perp)$). For every Boolean world $M : O \rightarrow \mathbb{B}$, write $M_1 = M|_{O_1}$ and let $\text{sel}(O, M) := \{\alpha \in O : M(\alpha) = \text{true}\}$ denote the sub-ontology selected by M . For nominal-free C, D in O_1 's signature,*

$$\text{Sat}(\text{sel}(O, M)) C \sqsubseteq D \iff \text{Sat}(\text{sel}(O_1, M_1)) C \sqsubseteq D.$$

That is, Sat-derivability of $C \sqsubseteq D$ at world M factors through the O_1 -restriction of M alone.

Proof. Two steps. (i) $\text{sel}(O, M)$ and $\text{sel}(O_1, M_1) \cup \text{sel}(O_2, M_2)$ are equal as ontologies (each axiom $\alpha \in O = O_1 \uplus O_2$ lies in exactly one side and contributes to the corresponding selected sub-ontology by the same Boolean choice $M(\alpha)$). Bidirectional sub-ontology inclusions yield `Sat` equivalence by monotonicity of `Sat` in the ontology (`Sat_mono`); the Lean witnesses are `ELKSDD.ELpp.selectedAxioms_sub_decompose` and `decompose_sub_selectedAxioms`. (ii) Apply Lemma 1 in the generalized form `ELKSDD.ELpp.Sat_factor_refined_general`, which uses O_1, O_2 as *signature-defining* outer ontologies while permitting the analyzed sides $\text{sel}(O_1, M_1)$ and $\text{sel}(O_2, M_2)$ to be sub-ontologies thereof. $\text{sel}(O_2, M_2)$ is consistent because it is a sub-ontology of the consistent O_2 , so the right disjunct of Lemma 1 vanishes, leaving the clean factorization through O_1 . See `ELKSDD.ELpp.per_world_sat_factor_consistent`. $\square$

Lemma 3 (Closed-form WMC SCC factorization under uniform priors).

Under the hypotheses of Lemma 2 (signature-disjoint, nominal-free, range-chain-safe O_1, O_2 with O_2 consistent), for the rational DISPONTE marginal under the uniform per-axiom weight $w \equiv 1$,

$$\text{wmc}^Q(O_1 \cup O_2, C, D, w) = \text{wmc}^Q(O_1, C, D, w) \cdot 2^{|O_2|}.$$

Proof. By definition,

$$\text{wmc}^{\mathbb{Q}}(O, C, D, w) := \sum_{M: O \rightarrow \mathbb{B}} [\text{Sat}(\text{sel}(O, M)) \ C \sqsubseteq D] \cdot \text{worldWeight}(M, w),$$

where the world weight is $\prod_{\alpha \in O} w(\alpha, M(\alpha))$. Three steps. (a) Under $w \equiv 1$, $\text{worldWeight}(M, w) = 1$ for every M (Lean: `worldWeightRat_uniform`). (b) By Lemma 2, the indicator depends on M only through $M_1 = M \upharpoonright O_1$. (c) The sum over $M : O_1 \uplus O_2 \rightarrow \mathbb{B}$ of any function depending only on $M \upharpoonright O_1$ factorizes as $\sum_{M_1: O_1 \rightarrow \mathbb{B}} f(M_1) \cdot 2^{|O_2|}$. Step (c) is a combinatorial identity proved by induction on $|O_2|$ over the flatMap structure of the world enumeration (ELKSDD.ELpp.sum_enumerateWorlds_factor); combining (a)–(c) yields the displayed equation. The closed-form Lean theorem is ELKSDD.ELpp.disponteWMCrat_uniform_scc_factor, audit-clean with dependencies only on [proptext, Classical.choice, Quot.sound]. $\square$

Lemma 4 (Distribution-semantics correspondence (rational form)).

For every $\mathcal{EL}^{++}$ ontology O , every concept pair (C, D) , and every per-axiom weight function $w : \text{DispAtom}(O) \times \mathbb{B} \rightarrow \mathbb{Q}$,

$$\text{wmc}^{\mathbb{Q}}(\text{compile}(O, C, D); w) = \text{wmc}^{\mathbb{Q}}(O, C, D, w),$$

where the LHS is the SDD-level WMC of the verified compiled circuit `compileSat` and the RHS is the rational DISPONTE distribution-semantics marginal. No restriction on w is imposed (in particular, no probabilistic normalization of w).

Proof. By Shannon-decomposition correctness of SDD compilation, the LHS equals the sum over all axiom-extensions of $[\text{Sat}(\text{sel}(O, M)) \ C \sqsubseteq D] \cdot \text{weightAlong}(M, w)$. The RHS is the same sum indexed by worlds. The two index sets coincide as multisets (a routine permutation lemma), and permutation-invariance of $\sum$ over $\mathbb{Q}$ closes the equality. This is the unconditional form: no *distributional assumption* is required; the entire Riguzzi 2015 distribution-semantics construction is internalized inside the SDD-WMC. Mechanized as ELKSDD.ELpp.wmc_compileSat_eq_disponteWMC_rat (Table 6). $\square$

D.9 DeepProbLog equivalence on the EL Horn theory

We tighten the informal claim of §5.3 that MOOSE on Γ_{EL} alone “yields the loss DeepProbLog [39] would minimize on the same Horn instance” into a formal proposition and a Lean 4 encoding in namespace `Moose` (the Lean identifiers retain the historical `no_clos` suffix as code artefacts).

Setup. Let $\mathcal{O}$ be an $\mathcal{EL}^{++}$ ontology, Δ a finite ABox domain, and let $\Gamma_{\text{EL}} = \text{AtomSub} \cup \text{Disj} \cup \text{Unsat} \cup \text{Links}$ be the ground clause set produced by the four MOOSE extractors after ELK saturation (Algorithm 1 of Section C, before `SDDCompile`). Let $\alpha_{\text{EL}} = \text{SDDCompile}(\Gamma_{\text{EL}})$, let $f_{\theta} : \mathcal{X}^{|\Delta|} \rightarrow [0, 1]^{|\mathcal{V}_{\sigma}^{\Delta}|}$ be a perception model assigning probability $p_{\theta}(\ell)$ to each positive ground literal (and weight $1 - p_{\theta}(\ell)$ to its negation), and let $\mathbf{e}$ be an evidence set of literals.

Translation $\mathcal{P}(\Gamma_{\text{EL}})$ to a ProbLog program. The translation maps Moose’s three Horn shapes plus the one non-Horn shape into the three native ProbLog constructs:

1. Each ground atom $A(a) \in \mathcal{V}_O^A$ becomes a *probabilistic fact* $p_\theta(A(a)) :: A(a)$.
2. Each definite Horn clause $\neg B_1 \vee \dots \vee \neg B_k \vee H \in \Gamma_{\text{EL}}$ (AtomSub, Unsat, and Links shapes) becomes a Datalog *rule* $H :- B_1, \dots, B_k$.
3. Each non-Horn atomic-disjointness clause $\neg A_1 \vee \dots \vee \neg A_n$ (Disj shape) becomes a ProbLog *integrity constraint* $:- A_1, \dots, A_n$.

Loss objects. $L_{\text{MOOSE/EL}}(\theta; \mathbf{x}, \mathbf{e}) := -\log \text{WMC}(\alpha_{\text{EL}}; p_\theta(\mathbf{x}), \mathbf{e})$ is the Semantic-loss WMC objective of §3.4 on the EL theory alone (no closure axioms); $L_{\text{DPL}}(\theta; \mathbf{x}, \mathbf{e}) := -\log P_{\text{DPL}}(\mathbf{e} \mid \mathbf{x}; \theta)$ is the DeepProbLog marginal loss [39] on $\mathcal{P}(\Gamma_{\text{EL}})$ under the ProbLog distribution semantics.

Proposition 1 (DeepProbLog equivalence on the EL Horn theory).
For every $(\mathbf{x}, \mathbf{e})$ consistent with $\mathcal{O}$,

$$L_{\text{MOOSE/EL}}(\theta; \mathbf{x}, \mathbf{e}) = L_{\text{DPL}}(\theta; \mathbf{x}, \mathbf{e}),$$

as values (in any commutative semiring carrying the literal weights) and as gradients in θ . The Lean mechanization instantiates the semiring at $\mathbb{N}$ (the type of `ELKSDD.SDD.wmc`); the same induction is semiring-polymorphic and lifts to $\mathbb{R}_{\geq 0}$ used in the Manhaeve 2018 real-valued marginal (paragraph (B) below).

Proof (Proof outline (full proof: `Moose.moose.no_clos_dpl_equiv` and `..._quantitative`)). Both losses equal $-\log \text{WMC}(\Gamma_{\text{EL}} \cup \mathbf{e}; w_\theta(\mathbf{x}))$, with w_θ assigning $p_\theta(\ell)$ to positive literals and $1 - p_\theta(\ell)$ to negative ones. SDD compilation preserves the model set, so $\text{WMC}(\alpha_{\text{EL}}; \cdot) = \text{WMC}(\Gamma_{\text{EL}}; \cdot)$ [21,17]. The translation $\mathcal{P}(\Gamma_{\text{EL}})$ defined above is shape-by-shape purely syntactic: a definite Horn rule, an atomic-disjointness integrity constraint, and a probabilistic fact each have the same satisfying assignments as the clause they came from, so $M \models \Gamma_{\text{EL}} \Leftrightarrow M \models \mathcal{P}(\Gamma_{\text{EL}})$. The DeepProbLog distribution semantics then coincides with WMC by [39, Eq. 4]: each total assignment over the probabilistic facts is weighted by $\prod_a p_\theta(a)^{M(a)} (1 - p_\theta(a))^{1-M(a)}$ exactly when it satisfies every rule and integrity constraint, which is the WMC weight of M . Equality of the losses as functions of θ implies equality of their gradients, and of any other operator respecting functional equality. $\square$

Lean encoding. The argument is mechanized in namespace `Moose` at two levels. At the qualitative level (`moose.no_clos_dpl_equiv`), both losses unfold to the same model-existence predicate $\exists M. M \models \Gamma_{\text{EL}} \wedge \text{support}(M) \subseteq \text{support}(w)$ and the equivalence closes by definitional unfolding (`Iff.rfl`) once the ProbLog translation is recognized as the identity on `List Clause`. At the quantitative level (`moose.no_clos_dpl_equiv_quantitative`), the equality is lifted from a proposition to a numeric identity `wmcQ O Δ w = dplLossQ O Δ w` via the key lemma `wmc.compileWithCtx.eq_modelSum`, which proves by structural induction on the variable list that the SDD weighted-model count equals the

explicit Manhaeve-Eq. 4 sum-over-models computed by Shannon decomposition. The proof is semiring-polymorphic; the Lean library instantiates the weighted-model-count semiring at $\mathbb{N}$ to keep the formalization free of a real-arithmetic dependency, while the same induction lifts verbatim to the $\mathbb{R}_{\geq 0}$ semiring of the original Manhaeve 2018 marginal. Gradient equality is a corollary (`moose_no_clos_dpl_loss_function_eq, ..._operator_invariant`): any operator respecting functional equality (differentiation, integration, point evaluation, PyTorch autograd) returns the same answer on both sides.

Reading. Proposition 1 pins MOOSE’s contribution to the EL-to-Horn compilation algorithm (Theorems 1 and 6): given $\mathcal{O}$ in $\mathcal{EL}^{++}$, the algorithm produces in time polynomial in $|\mathcal{O}|$ the ground Horn instance Γ_{EL} on which Semantic-loss WMC and the DeepProbLog loss coincide, lifting DeepProbLog to OWL 2 EL with existentials, role hierarchy, and role chains. The closure-augmented variant adds the disjointness, covering, and reverse-implication clauses of Φ_{clos} on top of Γ_{EL} . The content of these clauses is independent of $\mathcal{EL}^{++}$ entailment, and the DeepProbLog baseline correspondingly shows a higher RS-consistency rate (RS_{cons} , the rate of confident wrong commitments) than closure-augmented MOOSE across Tables 2 and 3.

D.10 Lean mechanization: theorem index

The Lean 4 library ELKSDD (introduced in Section A) formalizes every MOOSE-cited theorem of the OWL 2 EL stack; the companion library Moose houses the DeepProbLog-equivalence layer. Table 6 cross-references each paper claim with its namespace-qualified Lean theorem. Every theorem listed depends only on the standard Lean foundation [`propext`, `Classical.choice`, `Quot.sound`] (audit-clean), and the formalization contains zero `sorry` or `admit`. An audit pass in each library runs `#print axioms` on every paper-cited theorem at build time and reports its dependency set.

Table 6: Lean implementation index, namespace-qualified. Theorems are reported as `<namespace>.<theorem>`; the bulk live in `ELKSDD.ELpp`, with `ELKSDD.SDD` for the SDD calculus, `ELKSDD.RangeNorm` for the BBL 2008 path, and `Moose` for the DeepProbLog equivalence. Every listed theorem is audit-clean: `#print axioms` reports [`propext`, `Classical.choice`, `Quot.sound`] only, no MOOSE-specific axioms, no `sorry`, no `admit`. Each theorem stated in §4 corresponds to exactly one Lean theorem (top section); supporting lemmas, the closed-form WMC SCC factor under uniform priors (Theorem 4), and the LHS-nominal extension are listed in the per-topic sections that follow.

Paper claim	Lean theorem
<i>Section 4 (correctness and complexity): one Lean theorem per stated theorem</i>	
Verified SDD encoding (Theo-ELKSDD.ELpp.moose_inference_correct rem 1)	
models iff Sat	<code>ELKSDD.ELpp.compileSat_models_iff_sat</code>
size = $2^{ \mathcal{O} +1} - 1$	<code>ELKSDD.ELpp.compileSat_size_eq</code>
DISPONTE correspondence (Theorem 2)	<code>ELKSDD.ELpp.wmc_compileSat_eq_disponTEWMC_rat</code>
SCC compositional, Sat (Theo-ELKSDD.ELpp.scc_sat_factor rem 3)	

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Paper claim	Lean theorem
SCC compositional, WMC uni- form (Theorem 4)	ELKSDD.ELpp.disponteWMCrat.uniform.scc_factor
Reused ELK infrastructure (Section D.1; mechanized from [26])	
ELK soundness/completeness (Theorem 5)	ELKSDD.ELpp.correct_owl2el
soundness (unconditional)	ELKSDD.ELpp.sound_owl2el
completeness (with fragment witness)	ELKSDD.ELpp.complete_owl2el
Polynomial Sat decision (Theo- rem 6)	ELKSDD.ELpp.sat_decision_polynomial
exact closure-size bound	ELKSDD.ELpp.sat_closure_total_polynomial_bound
Closure-augmented circuit and inference complexity (appendix)	
Closure-augmented (Theorem 7)	ELKSDD.SDD.compile_correct
Inference complexity (Theo- rem 8)	ELKSDD.SDD.wmc_linear, ELKSDD.ELpp.compileSat.wmcCost_eq
Compositional and distributional results: supporting lemmas	
SCC Sat factor, sig-disjoint	ELKSDD.SCC.Sat_factor_refined
closed form (O_2 consistent)	ELKSDD.ELpp.scc_sat_factor, ELKSDD.ELpp.scc_sat_factor_symm
sub-ontology generalization	ELKSDD.ELpp.Sat_factor_refined_general
k-component generalization	ELKSDD.ELpp.scc_sat_factor_k, ELKSDD.ELpp.joint_consistent_pair
Per-world Sat factor (Lemma 2)	ELKSDD.ELpp.per_world_sat_factor_consistent
selectedAxioms decomposition	ELKSDD.ELpp.selectedAxioms_sub_decompose
World-summation factorization	ELKSDD.ELpp.sum_enumerateWorlds_factor
length-cast generalization	ELKSDD.ELpp.sum_enumerateWorlds_factor_general
WMC SCC factor (uniform, Lemma 3)	ELKSDD.ELpp.disponteWMCrat.uniform.scc_factor
DISPONTE corresp. (N-valued)	ELKSDD.ELpp.wmc_compileSat_eq_disponteWMC
DISPONTE corresp. (Q, Lemma 4)	ELKSDD.ELpp.wmc_compileSat_eq_disponteWMC_rat
existence form	ELKSDD.ELpp.exists_disponte_correspondence_rat
Nominal-aware SCC (LHS shape 1)	ELKSDD.ELpp.Sat_factor_nomLHS, ELKSDD.ELpp.scc_sat_factor_nomLHS
prodInterp on LHS-nominal axioms	ELKSDD.ELpp.prodInterp_satisfies_nomLHS
ELK calculus (Kazakov, Krötzsch, Šimánčík 2014)	
ELK soundness	ELKSDD.EL.sound, ELKSDD.ELpp.sound
ELK completeness (nom-free)	ELKSDD.ELpp.complete_via_canon
LHS-nominal extension	ELKSDD.ELpp.complete_via_canon_nomLHS
Canonical model $\models O$ (nom-free)	ELKSDD.ELpp.canon_satisfies
LHS-nominal extension	ELKSDD.ELpp.canon_satisfies_nomLHS
shallow- $\exists$ -nom-RHS extension	ELKSDD.ELpp.canon_satisfies_nomLR
Saturation termination	ELKSDD.ELpp.saturation_terminates
Sat $\Leftrightarrow$ derivable closure	ELKSDD.ELpp.sat_iff_in_derivableClosure
Poly-time Sat decision	ELKSDD.ELpp.sat_polynomial_decidable
Bounded Sat closure size	ELKSDD.ELpp.sat_closure_total_polynomial_bound
SDD calculus (Darwiche 2002, Choi–Darwiche 2013)	
SDD compile-correctness	ELKSDD.SDD.compile_correct, ELKSDD.SDD.compileWithCtx_correct
SDD WMC linearity	ELKSDD.SDD.wmc_linear, ELKSDD.SDD.wmcCost_eq_size
MOOSE pipeline (paper-citation theorems)	
Inference correctness	ELKSDD.ELpp.moose_inference_correct
Pipeline completeness	ELKSDD.ELpp.moose_pipeline_complete
Polynomial Sat decision (algo- rithmic)	ELKSDD.ELpp.sat_decision_polynomial
SCC summary	ELKSDD.ELpp.moose_scc_summary
SCC, chain-free corollary	ELKSDD.ELpp.scc_sat_factor_no_chain
SCC, range-free corollary	ELKSDD.ELpp.scc_sat_factor_no_range
Syntactic range-elimination [6] §3.3 (Path B)	
Syntactic range-elimination	ELKSDD.RangeNorm.eliminateRanges
strict-output preservation	ELKSDD.RangeNorm.eliminateRanges_strict
NoRange degenerate baseline	ELKSDD.RangeNorm.eliminateRanges_eq_under_no_range

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Paper claim	Lean theorem
rinc closure (bounded fixed point)	ELKSDD.RangeNorm.rincDescendants
transitive reflexive-rinc propagation	ELKSDD.RangeNorm.reflexiveTransitiveAxioms
Forward Sat-conservativity (full)	ELKSDD.RangeNorm.Sat_to_eliminated_full
transitive marker membership	ELKSDD.RangeNorm.reflexiveTransitive_axiom...
DeepProbLog equivalence (Manhaeve 2018, Riguzzi 2015)	
Qualitative (Proposition 1)	Moose.moose_no_clos_dpl_equiv
Quantitative ($\mathbb{N}$ -valued)	Moose.moose_no_clos_dpl_equiv_quantitative
WMC = model sum (BoolAtom)	Moose.wmc_compileWithCtx_eq_modelSum

E Inference, training, and RS-mitigation wrappers

This appendix gives the explicit inference equations of §3.4 (Section E.1), the gradient flow used at training time (Section E.2), and the MOOSE+BEARS / MOOSE+NeSyDM wrapper losses used in the RS-aware experiments (Section E.3).

E.1 Inference equations

The differentiable WMC layer supports three inference operations on the same SDD α , varying only literal weights and evidence.

Training loss. For $(\mathbf{x}, \mathbf{e})$,

$$L(\theta; \mathbf{x}, \mathbf{e}) = -\log \text{WMC}(\alpha; p_\theta(\mathbf{x}), \mathbf{e}). \quad (11)$$

The literal-weight map (Table 1) sets $w(\ell) := p_\theta(\ell)$ on positive ground literals $\ell \in \mathcal{V}_{\mathcal{O}}^{\Delta}$ and $w(-\ell) := 1 - p_\theta(\ell)$ on negative ones; evidence literals $\ell \in \mathbf{e}$ clamp the negation weight to zero (equivalent to conditioning the SDD on ℓ).

Conditional posterior at test time. For a query atom $C(a)$,

$$P_\theta(C(a) \mid \mathbf{x}, \mathbf{e}) = \frac{\text{WMC}(\alpha; p_\theta(\mathbf{x}), \mathbf{e} \cup \{C(a) := \text{true}\})}{\text{WMC}(\alpha; p_\theta(\mathbf{x}), \mathbf{e})}; \quad (12)$$

the latent-class prediction on a closed family $\mathcal{D}$ is the argmax $\hat{D}(a) := \arg \max_{D_i \in \mathcal{D}} P_\theta(D_i(a) \mid \mathbf{x}, \mathbf{e})$. The denominator is strictly positive whenever $\mathbf{e}$ is consistent with $\mathcal{O}$, so the ratio is well defined.

Perception-free entailment query.

$$\mathcal{O} \cup \mathbf{e} \models C(a) \iff \text{WMC}(\alpha; w_{1/2}, \mathbf{e} \cup \{C(a) := \text{false}\}) = 0, \quad (13)$$

where $w_{1/2}$ assigns $\frac{1}{2}$ to every literal: asserting $\neg C(a)$ leaves no $\mathcal{O}$ -consistent assignment iff the entailment holds. The same SDD acts as an off-line ELK reasoner over Δ , independent of any neural model. All three operations run in $O(|\alpha|)$ (Theorem 8).

E.2 Training step and gradient flow

A single training step on minibatch $\{(\mathbf{x}_i, \mathbf{e}_i)\}_i$ proceeds as follows.

1. *Perception forward.* The neural extractor $f_\theta : \mathcal{X}^{|\Delta|} \rightarrow [0, 1]^{|\mathcal{V}^\Delta|}$ maps the input tuple to a per-atom probability vector $p_\theta(\mathbf{x}_i)$, factorized as $\prod_\ell p_\theta(\ell \mid \mathbf{x}_i)$ (the marginal-independence factorization targeted by the RS literature, §2.2).
2. *Literal-weight assignment.* The atom-weight map of Table 1 routes $p_\theta(\ell \mid \mathbf{x}_i)$ to literal ℓ and $1 - p_\theta(\ell \mid \mathbf{x}_i)$ to $\neg\ell$. Evidence $\mathbf{e}_i$ clamps observed literals.
3. *SDD WMC traversal.* The post-order traversal visits each node once (memoized by node id) and accumulates the WMC value (Equation (8)) in $O(|\alpha|)$ time. Smoothness, decomposability, and determinism (Section D) make each sum disjoint and each product over disjoint scopes, so the result is exact.
4. *Loss aggregation.* $L(\theta) = -\frac{1}{n} \sum_i \log \text{WMC}(\alpha; p_\theta(\mathbf{x}_i), \mathbf{e}_i)$.
5. *Backward pass.* PyTorch autograd composes the backward pass through the same WMC traversal at $O(|\alpha|)$ additional cost, plus the perception backbone’s own backward pass. Every intermediate WMC value is the product/sum of literal weights, hence differentiable in θ via the chain rule: $\nabla_\theta L = -\frac{1}{n} \sum_i (\text{WMC}_i)^{-1} \nabla_\theta \text{WMC}_i$. No Monte-Carlo or sampling step is required; the gradient is computed exactly.

This exact-gradient property is what makes the SDD a tractable surrogate for the ProbLog marginal that DeepProbLog approximates by proof-tree enumeration (Proposition 1).

E.3 RS-mitigation wrapper losses

The compilation algorithm of §3.2 is independent of the perception’s output distribution: α depends only on $(\mathcal{O}, \Delta)$, so RS-aware methods plug in by replacing the literal weights $p_\theta(\mathbf{x})$ in Equation (11) without modifying α or the WMC traversal.

MOOSE+BEARS [40]. BEARS [40] trains a K -encoder ensemble $\{\theta_k\}_{k=1}^K$ that shares α but diversifies over the RS equivalence class. Members are trained sequentially: at step k , the new encoder θ_k is fit by maximizing the BEARS objective [40, Eq. 7]

$$L_{\text{BEARS}}(\theta_k; \mathbf{x}, \mathbf{e}) = L(\theta_k; \mathbf{x}, \mathbf{e}) + \gamma_1 \text{KL} \left(p_{\theta_k} \parallel \frac{1}{k} \sum_{j=1}^k p_{\theta_j} \right) + \gamma_2 H(p_{\theta_k}), \quad (14)$$

where the KL term against the running average over members $1, \dots, k$ encourages θ_k to commit to a different shortcut than its predecessors, and the entropy term $H(p_{\theta_k})$ prevents collapse to a degenerate posterior. At test time, the ensemble averages the per-atom marginals: $p(\ell \mid \mathbf{x}) = \frac{1}{K} \sum_k p_{\theta_k}(\ell \mid \mathbf{x})$, spreading mass across multiple shortcut explanations. We use $K=5$, entropy weight $\gamma_2=0.1$, and per-regime tuning of the KL weight γ_1 (Section F.4).

MOOSE+NeSyDM [30]. NeSyDM [30] replaces the marginal-independence factorization $p_\theta(\mathbf{c} \mid \mathbf{x}) = \prod_i p_\theta(c_i \mid \mathbf{x})$ with a masked-discrete-diffusion variational distribution $q_\theta(\mathbf{c} \mid \mathbf{x})$ that models cross-atom dependencies through a denoising trajectory over T timesteps. The continuous-time negative evidence lower bound (NELBO) objective is

$$\mathcal{L}_{\text{NeSyDM}}(\theta) = \mathbb{E}_{q_\theta}[-\log \text{WMC}(\alpha; q_\theta(\mathbf{x}), \mathbf{e})] + \gamma_c \mathcal{L}_c(\theta) + \gamma_h \mathcal{L}_h(\theta), \quad (15)$$

where $\mathcal{L}_c$ is the masked-diffusion concept-unmasking loss L_c of [30, §3.2] and $\mathcal{L}_h$ the variational-entropy term $L_{H[q]}$ that maximizes the entropy of q_θ at the fully-masked step. The expectation under q_θ is intractable in closed form; we use two gradient estimators: (a) *RLOO* [28], a leave-one-out REINFORCE baseline that draws S samples from q_θ and uses the sample average over the other $S-1$ as the control variate; (b) *exact-WMC*, which replaces sampling by direct WMC evaluation $-\log \text{WMC}(\alpha; \bar{q}_\theta, \mathbf{e})$ on the average per-atom marginal $\bar{q}_\theta$ when the diffusion-step posterior factorizes across atoms. We use the NeSyDM-repository defaults $\beta=10$ and $T=10$ continuous-time mask budget; (γ_c, γ_h) are swept per regime (Section F.4).

Both wrappers share α across all encoders/timesteps and inherit the $O(|\alpha|)$ inference cost of Theorem 8, multiplied by K for BEARS or by the diffusion budget $T \cdot S$ for NeSyDM.

E.4 Non-WMC baselines

LTN and ELEmbeddings replace exact WMC over the compiled SDD with, respectively, a fuzzy-logic satisfaction objective and a geometric embedding loss. Both consume the same atom vocabulary and per-example domain as the WMC-based methods (the same circuit atoms drive the losses), so the only thing varying across baselines is the symbolic treatment.

LTN [7]. Each $\mathcal{EL}^{++}$ axiom is grounded into a first-order formula over the per-example domain ($\{a\}$ for Exp. 1 / Track A–C, $\{a, b\}$ for Exps. 2–3) and scored under the product (Goguen) t-norm with Reichenbach implication ($a \wedge b = ab$, $a \vee b = a + b - ab$, $a \rightarrow b = 1 - a + ab$). The translations used are $C \sqsubseteq D \mapsto \forall x. C(x) \rightarrow D(x)$, $C \sqcap D \sqsubseteq E \mapsto \forall x. C(x) \wedge D(x) \rightarrow E(x)$, $C \sqsubseteq \exists R.D \mapsto \forall x. C(x) \rightarrow \exists y. R(x, y) \wedge D(y)$, and the symmetric variants for $\exists R.C \sqsubseteq D$, $C \sqcap D \sqsubseteq \perp$, $R \sqsubseteq S$, and $R_1 \circ R_2 \sqsubseteq S$. Per-atom truth values come from the same perception output that the WMC baselines use; $\forall$ and $\exists$ are aggregated by the generalized mean $\mathbf{p_mean}(v_1, \dots, v_n) = (\frac{1}{n} \sum_i v_i^p)^{1/p}$ with sweepable exponent $p \in \{2, 4\}$ (Section F.4). The training loss is $1 - \text{SAT}$ where SAT is the satisfaction aggregator across all axiom-grounded formulas. LTN does not compute an exact joint distribution over ground atoms.

ELEmbeddings [36, 54]. The same $\mathcal{EL}^{++}$ geometric encoding underlies DeepGO-SE [34], where ELEmbeddings approximate semantic entailment for protein-function prediction over the Gene Ontology; our use of ELEmbeddings here

adopts the same NF1–NF4 losses but attaches them to the MOOSE perception backbone rather than a sequence encoder. Each named concept C is embedded as an n -ball with centre $c_C \in \mathbb{R}^d$ and radius r_C ; each role R as a translation vector $\rho_R \in \mathbb{R}^d$. A linear head projects the perception’s CNN latent $\psi(\mathbf{x}) \in \mathbb{R}^{128}$ into the same $\mathbb{R}^d$ space, yielding a per-individual point z_a . The class-membership score is $P(C(a)) = \sigma(\text{margin} - (\|z_a - c_C\| - r_C))$ (image inside the ball $\Rightarrow$ atom true) and the role-atom score is $P(R(a, b)) = \sigma(\text{margin} - \|z_a + \rho_R - z_b\|)$. The TBox loss is the sum of mOWL’s NF1–NF4 losses [36,54] (mean-aggregated per normal form), and the total objective is $\mathcal{L} = \mathcal{L}_{\text{perception}} + \lambda_{\text{EL}} \mathcal{L}_{\text{TBox}}$ with $\mathcal{L}_{\text{perception}}$ a BCE over observed ABox literals. Vanilla EEmbeddings has no normal form for role chains, and role chains are ignored for EEmbeddings.

F Experimental details

This part of the appendix collects the materials supporting experiments: the full MNIST ontology used by all three MNIST experiments (Section F.2), per-method hyperparameter sweeps and the tuned configurations used in the result tables (Section F.4), and the full per-regime breakdown across all six metrics (Section F.6).

F.1 Compiled-circuit sizes and compilation cost

All experiments were run on a single machine with one NVIDIA RTX 4090 GPU and a single CPU core allocated per job. SDD compilation is CPU-only and happens once at start-up; the resulting circuit is reused across every training step, seed, and method, so its size is a deterministic function of the ontology $\mathcal{O}$ and the ABox domain Δ alone, not of the learning method. Every MOOSE variant and every SDD-based baseline in a given regime therefore shares the circuit in the corresponding row of Table 7.

Compile time grows with the ABox domain as analyzed in Section D.5: the SDD node count fits $\propto |\Delta|^{2.9}$ for $|\Delta| \leq 3$, with a $40\times$ compile-time jump at $|\Delta| = 4$. The $|\Delta| = 2$ configurations used throughout §5 sit well inside the sub-second regime.

Peak GPU memory at circuit-build time is 0 in every configuration: compilation is CPU-only and no tensor is allocated on the device before the encoder is moved there.

Training cost is dominated by the perception backbone rather than the WMC layer. On the MNIST regimes end-to-end training takes roughly 11 minutes at $|\Delta|=1$ (Exp. 1) and 65–75 minutes at $|\Delta|=2$ (Exps. 2–3), with peak GPU memory of 35–56 MB; the Pizzaiolo tracks train in under 9 minutes each (Track C, a single binary label, in well under a minute) but use ≈ 900 MB, reflecting the larger image encoder rather than any growth in the circuit (Table 7). Peak memory therefore tracks the backbone and batch size, not the compiled SDD, whose own device footprint is negligible.

Table 7. Compiled-circuit size and per-regime cost. SDD size (nodes, distinct atoms, CNF clauses) is method-independent; compile time is the range observed across all methods, learning rates, and seeds on one CPU core. Compilation finishes in ≤ 0.75 s in every configuration, and in ≤ 0.35 s in all but the two DeepProbLog cells (0.68 s on Exp. 2, 0.71 s on Exp. 3). *The Exp. 1 row was profiled separately on a CPU-only machine; its structural counts are machine-independent by the argument above, and its compile times are reported as measured there rather than merged into the RTX 4090 column. The last two columns report end-to-end training wall-clock and peak GPU memory during training for a single representative MOOSE-WMC run per regime (batch size 32; the MNIST CNN encoder on Exps. 1–3, the larger Pizzaiolo encoder on Tracks A–C); unlike the circuit columns these depend on the backbone, batch size, and epoch budget rather than on the compiled circuit.

Regime	$ \Delta $	SDD nodes	atoms	clauses	compile (s)	train (s)	peak GPU (MB)
MNIST Exp. 1 (atomic)*	1	155	14	117	0.18–0.24	647	35.3
MNIST Exp. 2 (relational)	2	1080	30	274	0.29–0.68	3893	56.4
MNIST Exp. 3 (role-chain)	2	1153	32	314	0.23–0.71	4484	56.4
Pizzaiolo Track A	2	245	26	103	0.16–0.30	516	903.4
Pizzaiolo Track B	2	245	26	107	0.14–0.23	507	903.4
Pizzaiolo Track C	2	16	6	10	0.15–0.23	26	903.3

F.2 Full MNIST ontology

The single OWL EL ontology $\mathcal{O}_{\text{MNIST}}$ used by all three MNIST experiments (§5.1) consists of 76 axioms over $\text{Sig}_C = \{D_0, \dots, D_9, \text{Even}, \text{Odd}, \text{Prime}, \text{Composite}\}$ and $\text{Sig}_R = \{\text{succ}, \text{plus_two}\}$, listed below in EL DL syntax. Numerals 0–9 are read as the corresponding D_i .

NF1: property subsumptions (18). $D_0 \sqsubseteq \text{Even}$; $D_2 \sqsubseteq \text{Even}$; $D_4 \sqsubseteq \text{Even}$; $D_6 \sqsubseteq \text{Even}$; $D_8 \sqsubseteq \text{Even}$; $D_1 \sqsubseteq \text{Odd}$; $D_3 \sqsubseteq \text{Odd}$; $D_5 \sqsubseteq \text{Odd}$; $D_7 \sqsubseteq \text{Odd}$; $D_9 \sqsubseteq \text{Odd}$; $D_2 \sqsubseteq \text{Prime}$; $D_3 \sqsubseteq \text{Prime}$; $D_5 \sqsubseteq \text{Prime}$; $D_7 \sqsubseteq \text{Prime}$; $D_4 \sqsubseteq \text{Composite}$; $D_6 \sqsubseteq \text{Composite}$; $D_8 \sqsubseteq \text{Composite}$; $D_9 \sqsubseteq \text{Composite}$.

NF2: disjointness (47). $\text{Even} \sqcap \text{Odd} \sqsubseteq \perp$; $\text{Prime} \sqcap \text{Composite} \sqsubseteq \perp$; plus $D_i \sqcap D_j \sqsubseteq \perp$ for every $0 \leq i < j \leq 9$ ($\binom{10}{2} = 45$ pairs).

NF3: successor existentials (10). $D_i \sqsubseteq \exists \text{succ}. D_{(i+1) \bmod 10}$ for each $i \in \{0, 1, \dots, 9\}$. The chain is cyclic ($D_9 \sqsubseteq \exists \text{succ}. D_0$); without the signature side condition (Section C) saturation would diverge.

NF7: role chain (1). $\text{succ} \circ \text{succ} \sqsubseteq \text{plus_two}$.

Saturation closure used by the experiments. ELK derives the chain consequence $D_i \sqsubseteq \exists \text{plus_two}. D_{(i+2) \bmod 10}$ via the role-composition rule $R_\circ$ (given $D_i \sqsubseteq \exists \text{succ}. D_{i+1}$, $D_{i+1} \sqsubseteq \exists \text{succ}. D_{i+2}$, and $\text{succ} \circ \text{succ} \sqsubseteq \text{plus_two}$, infer $D_i \sqsubseteq \exists \text{plus_two}. D_{i+2}$ [26]) populating $\mathcal{A}$ with the plus_two links that the role-chain experiment relies on. The four MOOSE extractors then ground Σ and $\mathcal{A}$ into the clauses of Γ_{EL} , and the closure axioms Φ_{clos} over the exhaustive family

$\{D_0, \dots, D_9\}$ supply $\binom{10}{2}m + m$ disjointness and covering clauses plus the six reverse clauses per individual (one per distinct parity-primality profile; §3.3).

F.3 Perception architecture

Both MNIST and Pizzaiolo experiments train the perception encoder from scratch. The MNIST encoder (used in Experiments 1–3) takes a 28×28 grayscale image through two Conv→ReLU→MaxPool(2×2) blocks of 32 and 64 channels (3×3 kernels, padding 1), flattens to $64 \cdot 7 \cdot 7$ features, and applies a fully-connected layer to a 128-dim latent followed by a linear head to per-concept logits. The Pizzaiolo encoder (Experiment 4) takes a 224×224 RGB image through four Conv→ReLU→MaxPool blocks of 32, 64, 128, 128 channels (3×3 kernels, padding 1), then global average pooling to 128 features and the same $128 \rightarrow 128 \rightarrow n_{\text{concepts}}$ head. Per-concept Bernoulli probabilities are obtained by a sigmoid; the same encoder is shared across all named individuals in a training instance.

F.4 Hyperparameter selection

For every method in the headline tables we ran a single-seed sweep over a small per-method grid, picked the best configuration per regime by RS_{cons} and Acc_{digit} on the seed-0 cell, and then ran a 5-seed final at that configuration. All methods share Adam [27] with batch size 32, 30 epochs on MNIST and 60 epochs on Pizzaiolo Tracks A/B (Track C uses 15 epochs because the binary `is_spicy` supervision saturates faster), unless otherwise noted. The reported MNIST headline tables in Tables 2 and 3 use 5 seeds at the tuned HPs; the Pizzaiolo tables (Tables 2 and 3) use 5 seeds at the tuned HPs except where indicated.

MOOSE / *Semantic-loss WMC*. Sweep over $lr \in \{10^{-3}, 3 \cdot 10^{-3}\}$ per regime. Best: mnist1 / mnist3 / pizzaC use $lr=3 \cdot 10^{-3}$; mnist2 / pizzaA / pizzaB use $lr=10^{-3}$. The SDD compilation is determined by $\mathcal{O}$ and Δ , so the only free knob is the perception backbone’s optimiser. A wider-range sensitivity analysis over this knob is in Section F.5.

DeepProbLog [39]. Sweep over $lr \in \{10^{-3}, 3 \cdot 10^{-3}\}$. Both regimes (mnist2, mnist3) prefer $lr=10^{-3}$; the `nn(·)` predicate’s softmax matches the latent signature Sig_C^ℓ exactly (one head per concept family); no additional HPs.

MOOSE+BEARS [40]. Ensemble size $K=5$, entropy weight $\gamma_2=0.1$ at the BEARS-paper default; sweep over the KL weight $\gamma_1 \in \{0.5, 1.0, 2.0\}$ on every regime. The MNIST sweep is uniform on $\gamma_1=1.0$ (the BEARS-paper default), so the final 5-seed run uses $\gamma_1=1.0$ on mnist1, mnist2, mnist3. On Pizzaiolo, γ_1 was tuned per track: pizzaA $\rightarrow \gamma_1=1.0$, pizzaB $\rightarrow \gamma_1=2.0$, pizzaC $\rightarrow \gamma_1=1.0$.

MOOSE+NeSyDM [30]. Sweep over $(\gamma_c, \gamma_h) \in \{0.1, 1.0\} \times \{1.0, 2.0\}$ per regime (4 cells), at the RLOO gradient estimator. $\beta=10$ and $T=10$ (continuous-time mask budget) are the NeSyDM-repository defaults, adopted unchanged. Best per regime: mnist1 $\rightarrow (\gamma_c=1.0, \gamma_h=2.0)$; mnist2 $\rightarrow (0.1, 2.0)$; mnist3 $\rightarrow (1.0, 1.0)$; pizzaA $\rightarrow (0.1, 2.0)$; pizzaB $\rightarrow (0.1, 2.0)$; pizzaC $\rightarrow (0.1, 1.0)$.

LTN [7]. Product T-norm; sweep over the p-mean aggregator exponent `forall_p=exists_p` $\in \{2, 4\}$ on every regime, and additionally over `exists_mode` $\in \{\text{universe}, \text{skolem}\}$ on MNIST relational/role-chain. Best: `forall_p=4` uniformly across all six regimes; `exists_mode=universe` on mnist2 and mnist3 (skolem produced equivalent or worse $\text{Acc}_{\text{digit}}$ at single seed). $lr=10^{-3}$.

ELEmbeddings [36] / *mOWL* [54]. Sweep over $(\text{embed_dim}, \text{margin}) \in \{50, 128\} \times \{0.1, 0.5\}$ per regime (4 cells). Best: mnist1 / mnist2 / mnist3 / pizzaB $\rightarrow (\text{embed_dim}=50, \text{margin}=0.1)$; pizzaA $\rightarrow (128, 0.5)$ (best Brier among the four cells; all four reached $\text{Acc}_C=1.0$ on the in-distribution split). For mnist3 the EL ball loss is augmented with an explicit `plus_two` role-chain penalty term (vanilla *ELEmbeddings* has no normal form for role chains; Section E.4).

Independent (BCE). Sweep over $lr \in \{3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}\}$ per MNIST regime. Best: mnist1 / mnist3 use $lr=3 \cdot 10^{-4}$ (best $\text{Acc}_{\text{atom}} + \text{ECE}$); mnist2 uses $lr=3 \cdot 10^{-3}$. Per-atom BCE on the observed evidence atoms only; no symbolic structure enters the loss.

Summary. The final numbers in Tables 2 and 3 are 5-seed means at the per-regime-tuned hyperparameters above, after a per-method single-seed sweep.

F.5 Learning-rate sensitivity

The per-method grids above are budget-matched: every method is tuned over a grid of the same size on the same seed-0 protocol, so no method is advantaged by a larger search. Because MOOSE exposes a single free hyperparameter, the perception backbone’s learning rate, while the SDD compilation has none, we additionally sweep it over a wider five-point grid to check that the headline numbers are not an artifact of a lucky rate. Table 8 reports MOOSE accuracy at each rate (3 seeds per cell); the metric is Acc_F on MNIST and Acc_C on Pizzaiolo (Section 5.3). Track C is not included in this sweep.

On MNIST and Pizza A the response is unimodal, with the optimum in the interior of the grid (10^{-3} or $3 \cdot 10^{-3}$) and the extreme rates worst, as expected; Pizza B is flatter and less regular, its per-rate means varying little (all within 0.786–0.846) with no clear interior peak. MOOSE is robust on Exp. 2 and Pizza A (spread ≤ 0.10) and more rate-sensitive on Exp. 3 and Pizza B (0.13–0.18), where the low-rate cells underfit and carry large seed variance. In every regime the tuned rate used for the headline tables sits at or adjacent to the best-performing rate in this grid, so the reported numbers are not a lucky-rate artifact.

Table 8. MOOSE learning-rate sensitivity (3 seeds per cell, mean $\pm$ s.d.). Metric: Acc_F on MNIST Experiments 2–3, Acc_C on Pizzaiolo Tracks A–B. The final row is the across-grid spread (max – min over all per-seed runs in the grid).

Learning rate	Exp. 2	Exp. 3	Pizza A	Pizza B
10^{-4}	0.729 ± 0.011	0.941 ± 0.055	0.861 ± 0.003	0.810 ± 0.056
$3 \cdot 10^{-4}$	0.744 ± 0.020	0.943 ± 0.068	0.906 ± 0.000	0.846 ± 0.010
10^{-3}	0.786 ± 0.030	0.943 ± 0.071	0.948 ± 0.007	0.827 ± 0.015
$3 \cdot 10^{-3}$	0.772 ± 0.022	0.994 ± 0.001	0.955 ± 0.000	0.786 ± 0.038
10^{-2}	0.743 ± 0.010	0.880 ± 0.077	0.925 ± 0.042	0.821 ± 0.044
spread (max–min)	0.100	0.178	0.096	0.129

F.6 Full per-regime results

Table 9: Full per-regime results, six metrics. Means $\pm$ s.d. over 5 seeds. The third column is Acc_F , family-argmax accuracy, on the MNIST blocks and Acc_C , per-atom accuracy on the latent slice, on the Pizzaiolo blocks, which declare no exhaustive family on Track C (Section F.7); the two are compared only within a block. Operative theory: $I_{\text{EL}} + \Phi_{\text{clos}}$ for the MOOSE rows on MNIST, I_{EL} for every other row (closure is opt-in and unused on Pizzaiolo), and none for Independent. $\text{RS}_{\text{cons}} = \text{Conf} \cdot (1 - \text{Acc})$ is the confidence-weighted error rate of that decode: lower is better, and a high value means confident commitment to a wrong latent explanation. F1_{macro} is the macro-averaged per-concept F1 on the latent signature. Bold marks the column-best within each block. On Track A the OOD harness records a single accuracy, so the Acc_{atom} and Acc_C columns coincide there by construction rather than by coincidence. DeepProbLog[†] rows on Pizzaiolo Tracks A/B are not separate runs: no ProbLog program was written for Pizzaiolo, and the cells reproduce the MOOSE (Semantic-loss WMC) values, which Proposition 1 shows the translation $\mathcal{P}(I_{\text{EL}})$ would attain exactly. Note that this translation uses one independent probabilistic fact per ground atom, whereas the MNIST DeepProbLog programs use an nn/4 annotated disjunction over the digit family, so the two halves of the row are different programs. RS_{cons} and F1_{macro} on Track C (binary is_{spicy} supervision) collapse to 0 across all methods because the metric is defined over a multi-class concept family; we mark these cells N/A. Pizzaiolo Track A $\text{RS}_{\text{cons}}/\text{F1}_{\text{macro}}$ are computed on the in-distribution eval block (the OOD harness does not record per-atom logits for these metrics).

Method	$\text{Acc}_{\text{atom}} \uparrow$	$\text{Acc}_{F/C} \uparrow$	$\text{NLL} \downarrow$	$\text{ECE} \downarrow$	$\text{RS}_{\text{cons}} \downarrow$	$\text{F1}_{\text{macro}} \uparrow$
<i>Experiment 1: atomic MNIST (RQ1). $\Delta =1$, NF1/NF2 fragment.</i>						
Independent (BCE)	58.0 ± 5.6	13.2 ± 7.9	2.9 ± 0.2	9.3 ± 4.5	15.7 ± 3.3	7.6 ± 5.7
DeepProbLog	89.0 ± 0.8	42.1 ± 1.3	1.2 ± 0.1	10.3 ± 0.9	31.3 ± 2.6	31.2 ± 6.9
LTN	92.2 ± 0.1	25.9 ± 6.5	1.1 ± 0.0	10.0 ± 0.0	47.1 ± 7.8	16.1 ± 6.7
ELEmbeddings/mOWL	90.0 ± 0.1	34.9 ± 4.2	0.3 ± 0.0	3.9 ± 0.4	14.9 ± 0.9	25.8 ± 2.8
MOOSE (Semantic-loss WMC)	89.3 ± 1.8	48.1 ± 10.1	1.6 ± 0.3	10.6 ± 1.8	29.0 ± 4.2	32.1 ± 10.0
MOOSE+BEARS	91.2 ± 0.9	50.1 ± 5.2	0.3 ± 0.1	3.9 ± 1.0	20.0 ± 1.4	28.0 ± 6.2
MOOSE+NeSyDM (RLOO)	90.6 ± 0.2	46.6 ± 2.1	0.3 ± 0.0	6.0 ± 0.6	23.2 ± 0.9	35.8 ± 1.9
MOOSE+NeSyDM (exact)	90.1 ± 1.6	49.4 ± 9.7	0.7 ± 0.2	5.1 ± 1.5	25.6 ± 2.1	35.9 ± 8.6
<i>Experiment 2: relational MNIST (RQ2). $\Delta =2$, NF3 cyclic-successor TBox.</i>						
Independent (BCE)	61.8 ± 4.9	8.8 ± 4.5	2.8 ± 0.3	13.4 ± 3.6	17.6 ± 2.3	5.7 ± 3.3
DeepProbLog	85.3 ± 0.8	38.9 ± 3.2	1.4 ± 0.1	12.3 ± 0.9	57.9 ± 3.2	17.5 ± 4.1
LTN	91.9 ± 0.1	14.5 ± 4.9	1.2 ± 0.0	10.0 ± 0.0	40.1 ± 8.6	11.4 ± 5.8
ELEmbeddings/mOWL	90.1 ± 0.2	34.8 ± 3.6	0.4 ± 0.0	7.2 ± 0.1	6.9 ± 0.4	25.8 ± 3.7
MOOSE (Semantic-loss WMC)	93.8 ± 0.3	74.6 ± 0.3	0.5 ± 0.1	4.9 ± 0.3	24.9 ± 0.3	63.7 ± 1.1
MOOSE+BEARS	93.9 ± 1.1	76.5 ± 5.9	0.2 ± 0.0	4.0 ± 1.2	18.0 ± 4.5	67.8 ± 9.3
MOOSE+NeSyDM (RLOO)	90.9 ± 0.4	58.2 ± 1.4	0.3 ± 0.0	7.3 ± 0.7	30.3 ± 0.8	45.8 ± 4.3
MOOSE+NeSyDM (exact)	92.5 ± 1.2	62.6 ± 6.1	1.2 ± 0.2	7.4 ± 1.2	37.2 ± 6.1	59.1 ± 4.5
<i>Experiment 3: role-chain MNIST (RQ3). $\Delta =2$, plus.two=succ $\circ$ succ.</i>						
Independent (BCE)	61.5 ± 3.3	12.2 ± 7.3	3.3 ± 0.4	9.0 ± 8.2	17.5 ± 3.5	7.9 ± 4.4
DeepProbLog	90.6 ± 3.1	59.6 ± 13.6	0.9 ± 0.3	8.0 ± 2.9	38.3 ± 12.7	48.4 ± 14.3
LTN	92.0 ± 0.1	11.0 ± 7.9	1.2 ± 0.0	10.0 ± 0.0	47.1 ± 6.9	8.2 ± 6.7
ELEmbeddings/mOWL	90.4 ± 0.1	27.4 ± 7.1	0.3 ± 0.0	6.3 ± 0.1	7.5 ± 0.7	18.7 ± 6.7

Continued on next page.

Table 9 continued.

Method	$\text{Acc}_{\text{atom}} \uparrow$	$\text{Acc}_{F/C} \uparrow$	$\text{NLL} \downarrow$	$\text{ECE} \downarrow$	$\text{RS}_{\text{cons}} \downarrow$	$\text{F1}_{\text{macro}} \uparrow$
Moose (Semantic-loss WMC)	98.7 $\pm$ 1.8	96.1 $\pm$ 7.2	0.1 $\pm$ 0.2	1.1 $\pm$ 1.5	3.9 $\pm$ 7.1	93.8 $\pm$ 8.6
Moose+BEARS	94.8 $\pm$ 0.5	89.2 $\pm$ 7.2	0.1 $\pm$ 0.0	5.6 $\pm$ 1.1	8.3 $\pm$ 5.2	81.0 $\pm$ 12.2
Moose+NeSyDM (RLOO)	91.8 $\pm$ 0.5	62.7 $\pm$ 2.9	0.2 $\pm$ 0.0	4.6 $\pm$ 0.8	28.0 $\pm$ 2.1	57.6 $\pm$ 2.0
Moose+NeSyDM (exact)	92.0 $\pm$ 1.2	60.1 $\pm$ 5.7	1.3 $\pm$ 0.2	7.9 $\pm$ 1.1	36.1 $\pm$ 9.7	62.4 $\pm$ 10.1
<i>Pizzaïolo Track A: pizza-type, OOD eval (RQ4). 5 held-out tie-breaker pizzas (Section 5.2).</i>						
Independent (BCE)	46.6 $\pm$ 6.2	21.7 $\pm$ 9.7	6.4 $\pm$ 5.0	58.3 $\pm$ 15.4	62.2 $\pm$ 12.3	11.1 $\pm$ 6.4
DeepProbLog [†]	68.1 $\pm$ 0.3	68.1 $\pm$ 0.3	4.1 $\pm$ 0.2	30.8 $\pm$ 0.7	4.2 $\pm$ 7.6	14.1 $\pm$ 3.5
LTN	74.0 $\pm$ 3.4	22.1 $\pm$ 5.2	13.6 $\pm$ 5.4	52.7 $\pm$ 21.8	57.6 $\pm$ 17.3	10.9 $\pm$ 6.8
ELEmbeddings/mOWL	77.0 $\pm$ 0.9	25.7 $\pm$ 7.3	4.9 $\pm$ 1.0	49.7 $\pm$ 10.2	54.9 $\pm$ 6.6	11.2 $\pm$ 6.5
Moose (Semantic-loss WMC)	68.1 $\pm$ 0.3	68.1 $\pm$ 0.3	4.1 $\pm$ 0.2	30.8 $\pm$ 0.7	4.2 $\pm$ 7.6	14.1 $\pm$ 3.5
Moose+BEARS	68.6 $\pm$ 0.3	68.6 $\pm$ 0.3	3.3 $\pm$ 0.3	28.2 $\pm$ 1.5	0.6 $\pm$ 0.7	30.1 $\pm$ 10.2
Moose+NeSyDM (RLOO)	72.2 $\pm$ 1.3	72.2 $\pm$ 1.3	0.6 $\pm$ 0.0	7.5 $\pm$ 1.2	34.3 $\pm$ 4.2	44.8 $\pm$ 6.3
Moose+NeSyDM (exact)	67.4 $\pm$ 1.4	67.4 $\pm$ 1.4	4.6 $\pm$ 0.8	30.8 $\pm$ 3.2	3.8 $\pm$ 8.0	43.4 $\pm$ 3.0
<i>Pizzaïolo Track B: property-class (RQ4, RQ5). 3-way RS over the property class.</i>						
Independent (BCE)	57.2 $\pm$ 3.3	25.0 $\pm$ 0.0	2.7 $\pm$ 0.9	26.4 $\pm$ 17.1	37.0 $\pm$ 12.8	12.0 $\pm$ 1.1
DeepProbLog [†]	82.4 $\pm$ 1.9	84.5 $\pm$ 2.1	2.4 $\pm$ 0.3	15.5 $\pm$ 2.1	49.8 $\pm$ 4.6	37.8 $\pm$ 1.1
LTN	83.3 $\pm$ 1.6	32.5 $\pm$ 20.9	7.0 $\pm$ 6.0	39.3 $\pm$ 17.4	28.8 $\pm$ 12.5	23.4 $\pm$ 19.0
ELEmbeddings/mOWL	76.9 $\pm$ 0.0	25.0 $\pm$ 0.0	0.6 $\pm$ 0.0	16.1 $\pm$ 0.2	19.5 $\pm$ 0.1	10.0 $\pm$ 0.0
Moose (Semantic-loss WMC)	82.4 $\pm$ 1.9	84.5 $\pm$ 2.1	2.4 $\pm$ 0.3	15.5 $\pm$ 2.1	49.8 $\pm$ 4.6	37.8 $\pm$ 1.1
Moose+BEARS	85.4 $\pm$ 0.8	87.9 $\pm$ 0.9	0.3 $\pm$ 0.0	12.3 $\pm$ 1.0	30.3 $\pm$ 8.6	42.0 $\pm$ 9.0
Moose+NeSyDM (RLOO)	80.1 $\pm$ 0.8	78.5 $\pm$ 0.8	0.5 $\pm$ 0.0	2.6 $\pm$ 0.9	38.0 $\pm$ 4.7	22.6 $\pm$ 2.3
Moose+NeSyDM (exact)	86.3 $\pm$ 1.9	84.7 $\pm$ 2.2	2.5 $\pm$ 0.4	15.3 $\pm$ 2.2	50.3 $\pm$ 4.4	37.7 $\pm$ 1.0
<i>Pizzaïolo Track C: is_spicy (RQ4, RQ5). Multi-witness disjunction; 15 epochs.</i>						
Independent (BCE)	55.2 $\pm$ 5.4	48.9 $\pm$ 5.6	1.0 $\pm$ 0.2	22.1 $\pm$ 9.2	N/A	N/A
DeepProbLog [†]	77.3 $\pm$ 2.8	81.0 $\pm$ 3.0	1.5 $\pm$ 0.3	15.3 $\pm$ 1.4	N/A	N/A
LTN	85.1 $\pm$ 0.9	84.8 $\pm$ 0.4	1.3 $\pm$ 0.3	14.1 $\pm$ 0.5	N/A	N/A
ELEmbeddings/mOWL	78.8 $\pm$ 6.5	75.4 $\pm$ 6.6	0.3 $\pm$ 0.1	11.8 $\pm$ 3.8	N/A	N/A
Moose (Semantic-loss WMC)	77.3 $\pm$ 2.8	81.0 $\pm$ 3.0	1.5 $\pm$ 0.3	15.3 $\pm$ 1.4	N/A	N/A
Moose+BEARS	79.9 $\pm$ 3.2	84.2 $\pm$ 3.6	0.4 $\pm$ 0.0	11.3 $\pm$ 3.4	N/A	N/A
Moose+NeSyDM (RLOO)	66.1 $\pm$ 3.7	63.9 $\pm$ 3.7	0.7 $\pm$ 0.0	8.9 $\pm$ 1.5	N/A	N/A
Moose+NeSyDM (exact)	75.9 $\pm$ 1.4	80.1 $\pm$ 1.5	0.8 $\pm$ 0.2	6.9 $\pm$ 2.2	N/A	N/A

Table 9 reports all six metrics (Acc_{atom} , the principal accuracy, NLL, ECE, RS_{cons} , F1_{macro}) per regime, laid out as six row-blocks (MNIST Experiments 1–3, Pizzaïolo Tracks A–C). The compact main-paper tables Table 2 and Table 3 are summary views of the principal-accuracy and ECE columns of this table; per-regime Acc_{atom} , NLL, RS_{cons} , and F1_{macro} appear only here. RS_{cons} and F1_{macro} on Pizzaïolo Track A are computed from the in-distribution eval block (the OOD harness writes only per-pizza accuracies); on Track C (is_spicy binary supervision) both metrics are defined over a multi-class concept family and are not applicable.

F.7 Family-argmax on Pizzaïolo

Family-argmax accuracy Acc_F is defined per individual as the argmax of the WMC posterior over a *declared exhaustive family*, scored against ground truth. On MNIST the digit family $\{D_0, \dots, D_9\}$ is declared exhaustive and Acc_F is the reported metric. Pizzaïolo does not admit the same decode across its three tracks; we therefore report Acc_C there and label it as such rather than folding two different quantities under one symbol. This appendix states the two metrics precisely and shows why the Pizzaïolo tracks carry Acc_C .

Track C is the clear case: is_spicy supervision is a multi-witness disjunction over four spicy toppings and declares no exhaustive family, so there is no set to take an argmax over and Acc_F is undefined; the harness records 0 for these cells by absence, not as a score, and we report them as N/A throughout. Track A is

Table 10. Pizzaiolo Track B under both accuracy metrics (5 seeds, mean $\pm$ s.d., %). Acc_F is the family-argmax decode over the 3-way property class; Acc_C is the per-atom accuracy on the latent slice reported in Table 2. Rows absent from the Acc_F column did not record a family decode. Tracks A and C are omitted: the metric is unavailable and undefined there respectively (see text).

Method	Acc_F	Acc_C
LTN	49.5 ± 3.6	83.2 ± 1.9
MOOSE (Semantic-loss WMC)	50.2 ± 4.6	84.5 ± 2.1
MOOSE+BEARS	49.8 ± 3.9	87.9 ± 0.9
MOOSE+NeSyDM (RLOO)	48.0 ± 5.3	78.5 ± 0.8
MOOSE+NeSyDM (exact)	49.7 ± 4.4	84.7 ± 2.2
spread (max-min)	2.2	9.4

reported on the out-of-distribution tie-breaker split, whose harness records per-pizza accuracies only and no family decode, so Acc_F is unavailable there without a rerun. Track B does admit the decode, over the 3-way property class, and we report it in Table 10: every method lands between 48.0 and 50.2, a 2.2-point spread against per-method standard deviations of 3.6–5.3, so the metric separates no pair of methods. The Acc_C column of the same runs spreads 78.5–87.9 and does separate them. Family-argmax on Pizzaiolo would therefore replace an informative metric with an uninformative one on the only track where it is even computable, which is why the headline tables keep Acc_C and name it.

F.8 Paired significance of the headline comparisons

A mean $\pm$ s.d. over five runs is a stability summary rather than a significance test, and the shared evaluation instances call for a paired analysis. We therefore rerun the principal comparisons over 20 seeds and test each MOOSE-vs-baseline difference with a paired Wilcoxon signed-rank test and an exact sign-flip permutation test on the mean difference, sharing seeds within every pair. Within each regime the p -values are Holm-corrected across the baselines, and we report significance at $\alpha = 0.05$. The metric is the regime’s headline accuracy: family-argmax accuracy Acc_F on MNIST Experiments 2–3, and latent-concept accuracy Acc_C on Pizzaiolo Tracks A–C (Table 11); the Pizzaiolo significance statements therefore concern the latent-atom metric.

The relational and role-chain regimes are unambiguous: MOOSE beats every propositional baseline on Experiments 2–3 by margins that are large and significant (Holm $p \leq 0.0002$). On Pizzaiolo the picture is mixed, and we report it as such. MOOSE is significantly stronger than NeSyDM on Tracks B and C, but BEARS is significantly stronger than MOOSE on both; on the Track A OOD split NeSyDM significantly overtakes MOOSE, while the MOOSE-vs-BEARS gap there is not significant. MOOSE is therefore competitive but not dominant under symbolic ambiguity, and its clear advantage is in the relational regime where EL-aware grounding propagates evidence that propositional encodings cannot.

Table 11. Paired significance of MOOSE (Semantic-loss WMC) against each baseline over 20 shared seeds. “diff” is the mean per-seed difference (MOOSE – baseline); “Holm p ” is the Holm-corrected Wilcoxon p -value within the regime. The verdict is *win* (MOOSE significantly higher), *loss* (significantly lower), or *ns* (not significant) at $\alpha = 0.05$. Metric: Acc_F on MNIST, Acc_C on Pizzaiolo. The 20-seed reruns cover MOOSE, DeepProbLog, NeSyDM (RLOO), LTN, and BEARS on Experiments 2–3 and Pizzaiolo Tracks A–C; Experiment 1 and the remaining baselines (Independent, ELEmbeddings, NeSyDM-exact) are 5-seed only and are not part of the paired analysis.

Regime	Baseline	MOOSE	base	diff	Holm p	verdict
MNIST Exp. 2	DeepProbLog	0.745	0.423	+0.322	0.0002	win
	NeSyDM	0.745	0.595	+0.150	0.0002	win
	LTN	0.745	0.150	+0.595	0.0000	win
MNIST Exp. 3	DeepProbLog	0.977	0.598	+0.378	0.0000	win
	NeSyDM	0.977	0.631	+0.346	0.0001	win
	LTN	0.977	0.151	+0.825	0.0000	win
Pizza A (OOD)	NeSyDM	0.679	0.709	−0.031	0.0004	loss
	BEARS	0.679	0.683	−0.004	0.1454	ns
Pizza B	NeSyDM	0.841	0.790	+0.052	0.0002	win
	BEARS	0.841	0.880	−0.038	0.0002	loss
Pizza C	NeSyDM	0.809	0.617	+0.192	0.0002	win
	BEARS	0.809	0.856	−0.047	0.0003	loss

The cell means here differ marginally from the 5-seed headline tables (Table 2) because this analysis uses the 20-seed reruns; e.g. MOOSE on Exp. 2 is 0.745 here versus 0.746 in the 5-seed table. These paired tests compare MOOSE against each baseline; the headline tables mark the raw column-best rather than a significance verdict, since the 20-seed reruns do not cover every method or regime (Table 11). The test covers accuracy only, so Table 3 still marks the raw column-best.

F.9 Inductive generalization

A further question is whether the learned models generalize to new query individuals rather than only to held-out images. We test this with an inductive held-out-edge split of the relational (Exp. 2) and role-chain (Exp. 3) regimes. On Exp. 2 we supervise only the successor edges whose source digit lies in $\{0, 2, 4, 6, 8\}$ ($0 \rightarrow 1, 2 \rightarrow 3, \dots, 8 \rightarrow 9$) and query individuals on the held-out edges $\{1, 3, 5, 7, 9\}$ ($1 \rightarrow 2, 3 \rightarrow 4, \dots, 9 \rightarrow 0$); on Exp. 3 the `plus_two` relation forms two 5-cycles, and we supervise sources $\{0, 1, 2, 3, 6, 7\}$ and query the held-out sources $\{4, 5, 8, 9\}$. In both splits the training and query edge sets jointly cover all ten digit nodes, so the shared perception network sees every digit during training; only the *relational configuration* of a query pair is novel, not its images; this is a test of relational generalization, not zero-shot perception. As a control we also evaluate on the supervised edge types with fresh images (the transductive column of Table 12).

MOOSE shows no inductive degradation on either regime: its held-out-configuration accuracy matches its transductive control to within noise (-0.032

Table 12. Inductive generalization over 20 seeds (Acc_F , family-argmax accuracy). *Inductive* evaluates on relational configurations never supervised; *transductive* is the same-edge-type control on new images; *drop* is transductive – inductive (positive means degradation on unseen configurations).

Regime	Method	Inductive	Transductive	Drop
Exp. 2 (relational)	MOOSE (WMC)	0.717 ± 0.071	0.685	−0.032
	Independent	0.582 ± 0.036	0.605	+0.023
	NeSyDM	0.135 ± 0.024	0.594	+0.459
Exp. 3 (role-chain)	MOOSE (WMC)	0.595 ± 0.138	0.574	−0.021
	Independent	0.551 ± 0.093	0.710	+0.159
	NeSyDM	0.254 ± 0.042	0.712	+0.458

and −0.021, i.e. marginally higher inductive). NeSyDM, by contrast, collapses from 0.594/0.712 transductive to 0.135/0.254 on held-out configurations, and the Independent baseline degrades moderately. The WMC layer scores whole declared families rather than memorized edge identities, so a query pair in an unseen configuration is handled by the same exact-inference path as a supervised one, which is what preserves accuracy. The paired MOOSE-vs-Independent advantage on the inductive split is significant on Exp. 2 (Wilcoxon $p < 10^{-4}$, $n = 20$) but not on Exp. 3 ($p = 0.52$), where the role-chain circuit leaves both methods with high seed-to-seed variance.

F.10 Closure ablation: separating Γ_{EL} from Φ_{clos}

The compiled theory combines the EL-aware extractors Γ_{EL} with the optional closure clauses Φ_{clos} , whose covering axiom $\top \sqsubseteq \bigsqcup_i D_i$ lies outside the EL profile. To isolate their contributions we disable Φ_{clos} on all three MNIST regimes (`--no-closed-world`, 5 seeds) and pair against the with-closure baseline (Table 13).

Table 13. Closure ablation on MNIST. Acc_F (mean $\pm$ s.d. over 5 seeds) with and without the exhaustive-family closure clauses Φ_{clos} . The **--no-closed-world** arm removes mutual exclusion and covering in all three regimes and profile-keyed reverse implications in Experiment 1. Δ is the change in the mean when Φ_{clos} is removed.

Regime	Method	with Φ_{clos}	without Φ_{clos}	Δ
Exp. 1 (atomic)	MOOSE	48.1 ± 10.1	21.7 ± 0.0	-26.4
	+BEARS	50.1 ± 5.2	21.7 ± 0.0	-28.4
	+NeSyDM (RLOO)	46.6 ± 2.1	21.7 ± 0.0	-24.9
	+NeSyDM (exact)	49.4 ± 9.7	21.7 ± 0.0	-27.7
Exp. 2 (relational)	MOOSE	74.6 ± 0.3	17.4 ± 1.8	-57.2
	+BEARS	76.5 ± 5.9	30.9 ± 4.5	-45.6
	+NeSyDM (RLOO)	58.2 ± 1.4	39.4 ± 3.0	-18.8
	+NeSyDM (exact)	62.6 ± 6.1	17.3 ± 5.1	-45.3
Exp. 3 (role-chain)	MOOSE	96.1 ± 7.2	9.4 ± 0.9	-86.7
	+BEARS	89.2 ± 7.2	37.1 ± 6.5	-52.1
	+NeSyDM (RLOO)	62.7 ± 2.9	46.5 ± 3.0	-16.2
	+NeSyDM (exact)	60.1 ± 5.7	54.3 ± 3.0	-5.8

Without Φ_{clos} the base WMC objective collapses toward chance in every regime, and the collapse deepens with relational complexity: Acc_F falls by 26.4, 57.2, and 86.7 points on Experiments 1–3, reaching 9.4 (chance is 10.0) on the role-chain regime. Experiment 1 is degenerate: all four methods land on the same seed-independent 21.7 ± 0.0 , because without the complete closure package the digit family receives no disambiguating gradient. The EL-aware extractors provide the structural scaffold, but the complete Φ_{clos} package supplies the learning signal that makes the latent digit identifiable; the MNIST gains cannot be attributed to I_{EL} alone.

The reasoning-shortcut mitigations can partially substitute for closure on the relational regimes, but which objective helps, and by how much, is regime-dependent rather than systematic. NeSyDM (RLOO) is the most robust, retaining 39.4 (Exp. 2) and 46.5 (Exp. 3) against plain MOOSE’s 17.4 and 9.4; BEARS buffers partially in both (30.9 and 37.1); and NeSyDM (exact) splits the two regimes, collapsing with the base objective on Experiment 2 (17.3, $\Delta = -45.3$) yet proving the most closure-robust of any method on Experiment 3 (54.3, $\Delta = -5.8$). The wrappers that resist collapse also hold RS_{cons} down (to 0.28–0.43, against plain MOOSE’s 0.43–0.50 where it does not collapse outright), reshaping the posterior enough to recover a substantial fraction of the signal, though never to the with-closure level. No single mitigation is a general substitute for Φ_{clos} : closure is necessary for the exact base objective and only partially, inconsistently replaceable by a reasoning-shortcut objective, which connects the mitigation behaviour of §5.3 to the I_{EL} -versus- Φ_{clos} separation.

G Related work: compilation, refinement, and embedding approaches

Rewriting-based approaches. $\mathcal{EL}^{++}$ is Datalog-rewritable, and a substantial line of work recasts description logic reasoning as rule evaluation: consequence-based calculi for EL can be expressed as Datalog programs [33], and data-independent transformations carry richer Horn DLs into Datalog while preserving assertion entailment [11]. MOOSE shares this view: Stage 2 of the pipeline (§3.2) treats the ELK saturation as a monotone Datalog program, but the two lines answer different questions. A rewriting produces a reasoning procedure whose output is an entailment set; it supplies no gradient and no per-individual marginal, so it cannot act as a supervision channel. The probabilistic description logics [12,13,24] attach a distribution to the axioms and answer probabilistic queries, but that distribution is specified by the modeller rather than learned from perception; the fuzzy line [53,8] gains differentiability by replacing model-theoretic entailment with a t-norm relaxation. MOOSE compiles the classical semantics into a circuit that is at once exact and differentiable, which is what allows neural outputs to serve as atom-level evidence.

Compilation-based approaches. Two recent works compile a description logic ontology into a differentiable representation for use with neural networks. Lazzari et al. [37] compile an $\mathcal{ALCI}$ TBox to a smooth, decomposable, deterministic circuit using a domino-style reduction, then plug the circuit into a multi-label classifier as either a Semantic-Loss regulariser or the head of a Semantic Probabilistic Layer; their experiments report on synthetically generated inputs and target the link-and classification-prediction setting in which every output label of the multi-label classifier is observed during training. Their treatment leaves three regimes open: the EL profile with role chains and role hierarchies, partial supervision in which a subset of the ground atoms is latent, and an RS analysis of the resulting predictor. DF- $\mathcal{EL}^{++}$ [53] approximates $\mathcal{EL}^{++}$ with a product-based fuzzy semantics that preserves PTIME-tractability and reports knowledge-base completion results on SNOMED CT (377K concepts); its loss replaces the classical model-theoretic entailment with a continuous relaxation, and it does not address exact WMC, partial-supervision concept learning, or RS metrics.

Refinement, clustering, and embeddings. Iterative Local Refinement (ILR) [19] corrects neural predictions at inference time so they satisfy a fuzzy-logic relaxation of a propositional formula; Embed2Sym [4] trains a perception network end-to-end on the downstream label, clusters the resulting embedding, and labels the clusters via a symbolic solver over a logic program. ELEmbeddings [36] maps concepts to n -balls and roles to translation vectors with NF1–NF4 margin losses that geometrize the $\mathcal{EL}^{++}$ axioms. DeepGOZero [35] trains a protein-sequence encoder jointly with these geometric class embeddings, supervising on observed protein–GO-class annotations via binary cross-entropy while the NF1–NF4 losses constrain the class n -balls through the GO axioms; predictions for a GO class with no training proteins are read off the same trained n -ball, giving zero-shot

annotations that ride on the ontology axioms alone. DeepGO-SE [34] reframes the same encoding as approximate semantic entailment. Our ELEmbeddings baseline (Section E.4) adopts the identical NF1–NF4 losses and ball-distance per-atom scoring but differs in the supervision regime, per-class function annotations there versus latent ABox atoms here. OWL2Vec* [15] embeds an ontology by training a word model on random walks over its axioms; a recent survey [16] covers the ontology-embedding landscape more broadly. ILR and Embed2Sym have not been instantiated for OWL EL with role chains; the embedding methods target subsumption inference and knowledge-base completion rather than partial-supervision concept learning. All four trade the model-theoretic semantics for a continuous representation, whereas MOOSE compiles the same ELK semantics into a differentiable circuit.